



**University of
Sheffield**

Assessing the Impact of Wind Renewable Energy on UK Grid Stability

An Empirical Analysis of Critical Thresholds and Machine Learning Approaches for Frequency Instability Prediction

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Abstract

The United Kingdom's electricity system underwent unprecedented transformation in 2024, achieving over 50% renewable generation while pursuing 95% clean power by 2030. Wind energy, comprising 42% of required capacity increases, emerges as critical to this transition, yet concerns persist regarding frequency instability events potentially intensifying with greater wind penetration. While theoretical studies propose various penetration limits, empirical validation using real-world UK operational data remains absent, creating significant knowledge gaps in understanding practical wind expansion constraints.

This study addresses three fundamental questions: how offshore versus onshore wind generation differentially influences grid frequency deviations, identification of critical wind penetration thresholds where stability metrics change significantly, and evaluation of machine learning approaches for forecasting grid instability events using historical operational data.

The methodology employs comprehensive ten-year data (March 2014-April 2024) from the National Energy System Operator and Open-Meteo services. Statistical analysis combines correlation assessment, Welch's ANOVA, and robust multiple regression to quantify wind impacts. Critical threshold identification utilises Binary Segmentation and Pruned Exact Linear Time changepoint detection algorithms with Bayesian Information Criterion validation. Predictive modelling implements ensemble methods incorporating Logistic Regression, Histogram-based Gradient Boosting, and LightGBM, evaluated through forward-chaining cross-validation.

Analysis reveals offshore wind demonstrates substantially stronger frequency deviation associations than onshore alternatives (Pearson correlations: 0.284 versus 0.097). Changepoint analysis identifies a statistically significant threshold at 25.5% wind penetration, beyond which instability events quadruple. The optimised ensemble achieves 33% recall and 31% precision, identifying one-third of instability events.

These findings provide the first comprehensive empirical validation of UK wind-grid stability relationships, offering operators concrete benchmarks for stability planning while demonstrating machine learning forecasting could generate £1.065 billion net present value, positioning predictive management as cost-effective alternative to capital-intensive infrastructure.

Keywords: wind energy, grid stability, frequency deviation, ensemble model, changepoint detection, UK grid system

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1. Introduction

The UK is currently in an extreme shift in the structure of the national grid, with 2024 seeing the closure of the final coal power plant and the first year with over 50% of generation from renewable energies (Department for Energy Security and Net Zero (DESNZ), 2025). This is a great step towards the UK's goal for clean power sources to meet 95% of generation and exceed 100% of the UK's demand by 2030 to become a net exporter of energy. There is still substantial progress to go, and with offshore and onshore wind energy together making up 42% of the required increase in installed capacity (DESNZ, 2024), wind will be the most significant indicator in meeting these goals. However, it has been suggested that meeting the 2030 target safely may be difficult due to the UK's insufficient preparation for frequency instability events which are expected to rise with increasing wind penetration (Fearn, 2025; Njoka et al., 2024).

The frequency of the national grid is a measure of the balance between the demand and generation of energy, and is set at 50Hz in the UK, at which the whole grid and all appliances are built to operate at (National Energy System Operator (NESO), 2025b). As the demand for energy increases or decreases, the UK's grid operator, NESO, ensures generation does the same to avoid frequency deviations beyond the ± 0.5 Hz statutory limit and the ± 0.2 Hz operational limit. If frequency deviates too much, generators trip offline to avoid damage and energy storage systems kick in to supplement the supply and avoid costly blackouts (Njoka et al., 2024). However, the current capacity of storage in the UK is not predicted to be sufficient to cope with the larger frequency deviations that occur under an increased proportion of wind in the energy mix (Mexis and Todeschini, 2020; Cárdenas et al., 2021), and there are no UK-specific studies on the extent we can increase wind utilising the current grid structure.

By combining real-world UK wind generation data with frequency records, this study aims to assess the national grid's tolerance to wind-related instability. In addition, we will evaluate the potential of machine learning models to act as early warning tools, ensuring a more secure and cost-effective approach to increasing the penetration of wind energy. Our findings are intended to inform operational thresholds, guide investment in stabilisation measures, and contribute to policy discussions around the economic and technical limits of wind expansion in the UK.

2. Scientific Context

Grid-Frequency Fundamentals and Inertia Loss

The UK electricity system operates at a target frequency of 50Hz with statutory limits of $\pm 1\%$ (49.5-50.5Hz) and tighter operational tolerances of $\pm 0.4\%$ (49.8-50.2Hz) that may only be exceeded for 1500 minutes annually (NESO, 2025b). Traditional synchronous generators provide frequency stability

through mechanical inertia, stored rotational energy that responds instantaneously to frequency deviations before control systems activate (Makolo et al., 2021). However, wind turbines connect via power electronic converters that decouple their rotational inertia from the grid, eliminating this natural stabilising effect (Johnson et al., 2020). This converter-connected generation fundamentally alters grid dynamics, increasing the rate of change of frequency (RoCoF) and reducing available response time during disturbances. NESO's Stability Pathfinder initiative has contracted 36 GVA.s of alternative inertia provision by 2026, yet this represents only 36% of minimum requirements (NESO, 2025b).

Documented Instability Events and Lessons

Three major blackouts demonstrate the vulnerabilities of low-inertia systems. South Australia's September 2016 blackout resulted from high wind penetration (40%) combined with low system inertia, where transmission faults triggered wind farm disconnections and cascading frequency collapse to below 47Hz (Australian Energy Market Operator Ltd (AEMO), 2017). Texas faced similar challenges in February 2021 when extreme weather exposed inadequate system resilience, affecting 4.5 million customers (Shrestha et al., 2023).

The UK's August 2019 event provides the most relevant precedent: near-simultaneous losses at Hornsea offshore wind farm (737MW) and Little Barford gas station (244MW) drove frequency to 48.8Hz within seconds, triggering automatic load shedding affecting one million customers (Ofgem, 2020). Critically, distributed generation losses exceeded 1.4GW as embedded systems disconnected to protect against perceived islanding conditions. These events highlight common failure modes, protective relay misoperation, insufficient inertia, and cascading disconnections, emphasising the importance of empirical UK thresholds rather than generalised limits.

Empirical vs. simulated wind-penetration thresholds

Literature reveals substantial variation in proposed wind penetration limits, largely dependent on methodology and system characteristics. Table 2.1 summarises key studies demonstrating this disparity.

Study	Grid Studied	Method	Threshold (%)	Key Limitations
Dai et al. (2019)	Generic system	Simulation	31	Idealised controls
Qing et al. (2020)	Chinese regional	Empirical/simulation	16-18	Small system scale
Qin et al. (2022)	Continental	Simulation	55-60	Perfect storage assumptions
Masood et al. (2015)	IEEE test system	Simulation	60	Fixed weather patterns
Jaghargh & Mashhadi (2020)	Generic	Simulation	45	Simplified grid response

Table 2.1: Wind Penetration Threshold Studies

Most studies rely on simulated environments that overestimate practical limits through simplified control systems, perfect ramping assumptions, and idealised weather patterns (Ahmed et al., 2023). Only Qing et al. (2020) incorporated historical frequency data, while others used controlled scenarios that fail to capture real-world complexity. No study addresses UK-specific seasonal wind patterns or the unique characteristics of the GB system's island operation. Recent empirical evidence suggests simulated thresholds consistently exceed achievable penetration levels due to unmodelled system interactions and protection scheme responses.

Onshore Vs. Offshore Wind Impacts

Offshore wind demonstrates superior consistency with capacity factors 15-20% higher than onshore equivalents, attributed to stronger, more consistent wind resources and reduced wake effects (Tumse et al., 2024). However, offshore installations typically connect via HVDC transmission systems that introduce 10-50ms response delays during frequency events (Rahman et al., 2021).

While advanced Voltage Source Converter (VSC) technology can provide faster response times, implementation strategies remain under development (Koondhar et al., 2023). UK offshore wind capacity reached 14.7GW in 2024, with projects like Eastern Link (2GW, commissioned 2029) further increasing HVDC penetration. Onshore wind, despite higher variability, provides sub-5ms response times due to direct AC connection to demand centres (Foody et al., 2024; RenewableUK, 2024). The March 2021 wind drought, when UK wind operated at just 11% capacity, highlighted vulnerabilities regardless of technology type (Potisomporn, 2023). Critically, no empirical comparison exists of actual UK frequency deviations attributable to onshore versus offshore wind performance under real operating conditions.

Economics and Regulation Of Stability Services

As system inertia declines and frequency instability increases, the National Electricity System Operator (NESO) faces rising balancing costs, reaching £2.4 billion last year (NESO, 2024). However, this figure is not transparently weather-adjusted or clearly defined in terms of methodology, and Ofgem's estimates may differ (Ofgem, 2024). These costs are passed to consumers via higher tariffs (NESO, 2024), emphasising the need for cost stabilisation.

Current initiatives like the Stability Pathfinder aim to secure alternative inertia sources, but contracted volumes remain below requirements (NESO, 2025b). The UK's Capacity Market now includes inertial products, yet their cost-effectiveness compared to international benchmarks and other technologies (e.g., batteries, synchronous condensers) is underexplored (Makolo et al., 2021; Johnson et al., 2020).

Improving forecasting accuracy is widely recommended to reduce balancing expenses by enabling proactive management (Chen et al., 2024; International Energy Agency, 2022). Policymakers should

clarify cost methodologies, compare international approaches, and consider academic cost-benefit analyses to justify further wind penetration or alternative solutions.

Machine-Learning for Early-Warning Tools

Machine-learning has already boosted wind-power forecasting accuracy, yet its use for predicting grid-frequency stability is still in its infancy. Tree-based ensembles such as XGBoost reliably convert meteorological inputs to power output while preserving a degree of interpretability (Wang et al., 2024). For the heavily imbalanced datasets typical of frequency-deviation studies, the Precision–Recall AUC is a more informative metric than overall accuracy or ROC-AUC (Saito & Rehmsmeier, 2015). A persistent drawback, however, is that averaging across many trees dampens the distribution’s tails, causing rare but critical excursions to be under-predicted (Hallgren et al., 2021). Weighted k-Nearest Neighbour models that embed generation-mix features have recently shown stronger performance on >10-minute horizons (Onsaker et al., 2023). To date, none of these approaches has been trained or validated on the extensive UK-specific wind-generation and frequency data sets, leaving a significant gap in real-time stability-assessment capability for Great Britain.

Synthesis

The literature reveals three fundamental gaps constraining UK grid stability assessment under rising wind penetration.

- **Empirical impact:** no study isolates the respective effects of offshore and onshore wind on second-by-second frequency, so operators lack technology-specific insight.
- **Penetration thresholds:** the excursion rate’s break-points, and their seasonal or diurnal drift, have never been quantified, leaving inertia and FFR requirements sized by broad heuristics rather than data-driven limits.
- **Predictive tools:** machine-learning models that could anticipate instability events remain untested on the rich UK dataset, forcing ESO to buy reserves conservatively and inflating balancing costs.

These gaps motivate empirical investigation of UK wind generation impacts on frequency stability, establishment of evidence-based penetration thresholds, and development of ML-based early warning systems using historical GB system data.

To address these gaps, this study investigates the following research questions:

1. How do offshore and onshore wind generation, individually and combined, influence grid frequency deviations from 50 Hz in the Great Britain system?

2. What are the wind penetration levels at which significant changes in grid stability metrics occur, and how do these thresholds vary seasonally and diurnally?
3. Can machine learning approaches effectively forecast binary grid instability events using historical data on wind generation, demand, and weather in the UK?

3. Data Acquisition and Pre-processing

To investigate the relationship between wind energy penetration and grid stability in the UK, we required high-resolution data (from 01/03/2014 00:00:00 to 01/04/2024 00:00:00) spanning across four key aspects of the power system: System frequency, generation mix, meteorological information and national demand data. This time frame was chosen to ensure we captured these rare instability events and the rapid growth in offshore wind capacity (Shen et al., 2024).

We reviewed UK energy system operator’s public repositories, they were evaluated against data granularity, collection reliability, accessibility and completeness of metadata documentation. NESO was chosen for system frequency; generation mix and national demand; Open-Meteo was chosen for meteorological data. These two were chosen as they were the most accessible repositories, both had API capabilities, and little data quality or accessibility issues were faced when acquiring their data.

Preprocessing involved aligning all datasets to hourly resolution to match the weather data. Frequency, generation, and demand were aggregated using means (for clarity) and linear interpolation was used to handle gaps effected by daylight saving times. Outliers were identified as those outside $Q1-1.5IQR$ and $Q3+1.5IQR$, due to its robustness to non-normal distributions (see Appendix 1.2), and they were retained for analysis as they reflect meaningful system instability or contributing conditions.

Feature engineering focused on capturing frequency deviations to be used in all our research questions (Table 3.2). We used the half-hourly wind energy penetration data and quarterly onshore/offshore ratios (DESNZ, 2024) to calculate half-hourly onshore/offshore splits (see Appendix 1.1).

The final pre-processed and engineered dataset is summarised in Tables 3.1–3.6 below:

Feature	Description
nd	National demand in MWh
nd_lag_1/2/3	For time series analysis and forecasting: reading 1/2/3 hours before the prediction point

Table 3.1: National Demand

Feature	Description
mean_frequency	The mean system frequency
standard_deviation	The standard deviation of the system frequency
min_negative_deviation	Largest system frequency drops below 50Hz
max_positive_deviation	Largest system frequency rise above 50Hz
mean_abs_deviation	The mean of absolute differences from 50Hz
max_abs_deviation	The largest absolute difference from 50Hz
minutes_outside_normal	Minutes spent outside the range of ± 0.2 Hz
instability_event	Binary indicator (1 if minutes_outside_normal > 5 or max_abs_deviation > 0.3, else 0)
stability_category	Categorical classification of stability based on max_abs_deviation ('stable': < 0.1, 'moderate': 0.1-0.2, 'concerning': 0.2-0.3, 'severe': > 0.3)
mean_abs_deviation_lag_1/2/3	For time series analysis and forecasting: reading 1/2/3 hours before the prediction point
max_abs_deviation_lag_1/2/3	For time series analysis and forecasting: reading 1/2/3 hours before the prediction point
minutes_outside_normal_lag_1/2/3	For time series analysis and forecasting: reading 1/2/3 hours before the prediction point

Table 3.2: System Frequency Features

Feature	Description
season	Categorised as Winter (Dec-Feb), Spring (Mar-May), Summer (Jun-Aug), Autumn (Sep-Nov)
time_of_day	Categorised as Night (0-6), Morning (6-12), Afternoon (12-18), Evening (18-24)
is_weekend	Binary flag (1 for Saturday/Sunday, 0 for weekdays)
demand_period	Categorised as Morning_Peak (7-9), Evening_Peak (16-19), Off_Peak (23-6), and Standard (all other hours)

Table 3.3: Temporal and demand pattern features

Feature	Description
wind	Wind generation (excluding embedded)
wind_emb	Embedded wind generation
total_wind	Total wind generation
onshore_wind	Generation from onshore wind farms
offshore_wind	Generation from offshore wind farms
onshore_penetration	Percentage of total generation from onshore wind ($\text{onshore_wind} / \text{total_generation} * 100$)
offshore_penetration	Percentage of total generation from offshore wind ($\text{offshore_wind} / \text{total_generation} * 100$)
wind_penetration	Percentage of total generation from wind ($\text{total_wind} / \text{total_generation} * 100$)
onshore_penetration_lag_1/2/3	For time series analysis and forecasting: reading 1/2/3 hours before the prediction point
offshore_penetration_lag_1/2/3	For time series analysis and forecasting: reading 1/2/3 hours before the prediction point

Table 3.4: Wind Generation Features

Feature	Description
temperature_2m	Air temperature at 2m height
relative_humidity_2m	Relative humidity at 2m
dew_point_2m	Dew point temperature at 2m
pressure_msl	Atmospheric pressure at mean sea level
wind_direction_10m	Wind direction at 10m height
wind_direction_100m	Wind direction at 100m height
wind_gust_10m	Wind gust speeds at 10m height
wind_speed_10m	Wind speed at 10m height
wind_speed_100m	Wind speed at 100m height
storm_indicator	Binary flag for high wind gust periods (1 if $\text{wind_gusts_10m} > 20$ m/s, else 0)

Table 3.5: Weather Data Features

Feature	Description
gas	Gas generation in MW
coal	Coal generation in MW
nuclear	Nuclear generation in MW
hydro	Hydro generation in MW
imports	Imports in MW
biomass	Biomass in MW
other	Other in MW
solar	Solar in MW
storage	Storage in MW
total_generation	Sum of all generation sources (gas, coal, nuclear, hydro, imports, biomass, other, solar, storage, wind, wind_emb)
synch_gen_share	Proportion of generation from synchronous sources that provide physical grid inertia, formula: $(\text{gas} + \text{coal} + \text{nuclear} + \text{hydro} + \text{biomass}) / \text{total_generation}$

Table 3.6: Generation Mix Features

In the NESO dataset we identified reporting quality issues from remote regions in Scotland, potentially underrepresenting areas with significant wind capacity. The weather data from the Open-Meteo exhibited similar disparities, with higher uncertainty in the coastal regions where many wind farms are located. This raises concerns about the fairness of our analysis, as conclusions may be biased towards regions with more accurate data. After a thorough evaluation, alternative sources were not an option due to limited accessibility and availability. These reporting quality challenges could affect our third research question, as a model is as strong as its training data, so it could result in a less accurate predictive model.

Both NESO and Open-Meteo maintain quality assurance processes; NESO adhere to regulatory reporting requirements (NESO, 2025c) and undergoes continuous validation against multiple measurement points (NESO, 2016).

Open-Meteo weather data is derived from multiple meteorological models with published validation statistics, allowing us to account for forecast uncertainty in our analysis (Open-Meteo, 2022). The use of open data sources ensures reproducibility from the outset, and both sources provide metadata, measurement methodologies and interpolation methods (NESO, 2025a; Open-Meteo, 2022).

We additionally considered the ethical risk where generation patterns reveal commercially sensitive operation strategies for specific plants, but focusing on system-level patterns rather than individual generator behaviours eliminates this. The central concern surrounds the safety ramifications of these issues. Data quality challenges could lead to misidentification of critical stability thresholds, with implications for grid operation recommendations.

4. Methodology and Planning

4.1 Planning

Our twelve-week schedule (see Gantt chart and timeline in Appendix 2.1) runs from the brief release on 26 Feb 2025 to report submission on 22 May 2025 and is built around three gates (Table 4.1):

Phase	Weeks	Gate / deliverable
Project definition – topic choice, architecture, WBS	1–2	Scope locked 7 Mar
Initial analysis and poster – rapid answers to the main RQs, figures for poster	3–9	Poster draft 1 May → poster submitted 6 May
Extended analysis and report – deep statistics, economic tests, modelling, write-up	9–12	Full analysis 19 May → final report 22 May

Table 4.1: Project phases, timeline and key deliverable gates

Roles and Dependencies:

- **Project management** – Henok
- **Data collection and preprocessing** – Adam (Henok) → kicks off 21 Mar; supplies all streams.
- **Quick statistical and economic checks** – Joaquin and Jori → run on early extracts (15 Apr – 1 May) for poster.
- **Poster assembly** – All, led by Adam (Jori) → integrate visuals and text (25 Apr – 6 May).
- **Deep statistical analysis** – Joaquin → continues after poster (6 May – 19 May).
- **Economic analysis** – Jori → parallel with deep stats (6 May – 19 May).
- **Feature engineering and model tuning** – Amir (Henok) → starts with first data cut, intensifies post-poster (4 Apr – 19 May).
- **Literature review and editing** – Catrin → ongoing, merges result into report (13 Mar – 19 May).

Risk Register

We maintain a living risk register that is reviewed at every Thursday and/or Friday stand-up to keep threats visible and actionable. The current high-level entries, with assessed likelihood, impact and planned responses, are summarised below:

Risk	Likelihood	Impact	Mitigation / Contingency
Data ingestion delays	Medium	High	Begin parallel feature engineering on existing extracts; request expedited access
Compute resource shortages	Low	Medium	Prototype on smaller samples; schedule compute jobs off-peak; leverage cloud burst
Model overfitting / poor generalisation	Medium	High	Reserve an extra validation fold; simplify models; increase regularisation; augment data
Library / version incompatibilities	Low	Low	Lock environments via requirements.txt / conda.yml; test upgrades on a dev branch
Timeline slippage in CV experiments	Medium	Medium	Reduce grid-search granularity; parallelise fold runs; build in a one-week buffer

Table 4.2: Risk register with likelihood, impact and mitigation actions

4.2 Methodology

4.2.1 Statistical Testing

To address the first research question, which investigates how the share of offshore versus onshore wind generation affects the size of frequency deviations in the UK power grid, we employed a mixed-method statistical framework that combines group-based inference with multivariable regression.

The aim was to identify whether systematic differences exist in frequency deviation metrics across varying wind penetration levels and to quantify their effects after adjusting for contextual factors such as season and time of day.

We began by formulating hypotheses. The null hypothesis (H_0) states that there is no significant difference in frequency deviations across different levels of offshore and onshore wind penetration. The alternative hypothesis (H_1) suggests that at least one level of offshore or onshore wind penetration is associated with a significantly different deviation.

The analytical workflow consisted of several key stages:

1. Correlation assessment using Pearson and Spearman methods to evaluate the direction and strength of monotonic and linear relationships between wind penetration variables and deviation measures.
2. Normality and variance diagnostics, including Shapiro-Wilk tests and Levene's test, to inform the use of parametric or non-parametric methods.
3. Binning of wind penetration into 5% intervals to support group-based comparisons.
4. Welch's ANOVA and Games-Howell post-hoc tests were used to detect statistically significant differences between penetration levels.
5. Construction of a multiple linear regression model incorporating onshore and offshore wind shares, their interaction, and control variables (wind speed, season, time of day, weekend effect).
6. Heteroscedasticity diagnostics using Breusch-Pagan and White's tests assessed violations of constant variance assumptions and motivated the use of robust standard errors.

To ensure robust inference, we applied HC3 robust standard errors, which are specifically designed to adjust for heteroscedasticity in linear models and are recommended for small to moderate sample sizes due to their tendency to reduce bias in standard error estimates (Long & Ervin, 2000). Multicollinearity was assessed using Variance Inflation Factors (VIFs), and final results were supplemented by seasonal and time-of-day bar plots to highlight contextual variation. This methodological structure ensured both the detection of group-level differences and the estimation of marginal effects under realistic operational conditions.

4.2.2 Threshold Analysis

To answer the second research question, which explores whether there is a critical level of wind penetration at which frequency instability increases significantly, we implemented a changepoint detection strategy grounded in non-parametric estimation and model selection.

Unlike RQ1, which focuses on continuous associations and group comparisons, RQ2 investigates whether a structural shift or threshold effect exists in the relationship between wind penetration and frequency instability, defined as the number of minutes per hour that the grid falls outside the ± 0.2 Hz operational band.

The null hypothesis (H_0) states that there is no changepoint in this relationship; any increase in instability occurs gradually or linearly. The alternative hypothesis (H_1) suggests that a discrete changepoint exists, and this threshold may differ across seasons.

The methodology followed six major components:

1. Correlation screening using Pearson and Spearman coefficients to evaluate whether any linear or monotonic trend existed.
2. Distribution testing of the instability metric using Shapiro-Wilk to confirm the need for non-parametric techniques.
3. Binning of wind penetration in 1% increments and estimation of bootstrapped mean instability and 95% confidence intervals per bin.
4. Binary Segmentation (BinSeg) and PELT algorithms were applied to detect the most statistically justified changepoint. BinSeg is a greedy optimisation technique that partitions a time series by iteratively locating the most significant changepoints (Kovács et al. 2022), while PELT (Pruned Exact Linear Time) optimises changepoint detection with a pruning strategy to achieve computational efficiency and exact minimisation of cost functions (Killick et al., 2012).
5. Applied two minimum data sufficiency rules to reduce the risk of overfitting. First, each 1% penetration bin was required to contain at least 30 hourly observations to ensure stability in bootstrapped estimates, which is commonly used in statistical sampling due to its foundation in the Central Limit Theorem (Chernick & LaBudde, 2011; Hesterberg, 2015). Second, any model split, such as a changepoint, was only considered valid if it resulted in at least five usable bins on either side of the breakpoint, a conservative rule of thumb used to maintain statistical power in changepoint segmentation.
6. Our Threshold validation uses the Bayesian Information Criterion (BIC), a likelihood-based model selection metric that incorporates a penalty for model complexity (Drton & Plummer, 2017; Bickel, 2025). We decided to apply a minimum BIC improvement threshold of $\Delta\text{BIC} \geq 6$ to determine whether introducing a changepoint significantly improves model fit.

Each changepoint was reported using the bin midpoint for clarity and consistency, and separate models were estimated for each season and time-of-day grouping to examine potential temporal variation. Visual line plots annotated with bootstrapped confidence intervals and changepoint indicators were used to interpret results. This methodology allowed us to identify robust, interpretable thresholds under varying operational conditions and provided insight into when grid instability begins to rise significantly.

4.2.3 Predictive Modelling

To address the third research question, whether machine learning approaches can forecast binary grid-instability events using historical data, we adopted a systematic multi-model evaluation approach. Our methodology focused on robust evaluation through temporally aware cross-validation and ensemble

techniques to overcome challenges specific to imbalanced time-series classification (Ganaie et al., 2022).

The methodological workflow comprised five key stages:

1. Feature Engineering

We created domain-specific features capturing temporal patterns (cyclical hour encoding, season, time-of-day), rolling statistics (3-hour wind averages, 6-hour deviation volatility), and derived measurements (distance from critical thresholds, pressure trends). This approach follows best practices for energy system modelling as recommended by Fodstad et al. (2022).

In total, 25 candidate features were created to capture the multidimensional dynamics of grid stability (see Appendix 2.2.3).

2. Feature Selection

We first filtered the candidate predictors with a Variance Inflation Factor (VIF) screen to eliminate deterministic or highly collinear variables and then ran a stability-based feature-selection procedure using forward-chaining cross-validation with annual folds from March-to-March (2015-2022) (see Appendix 2.4.2.1).

This two-stage process respects the chronological ordering of observations, controls redundancy, and ultimately reduces dimensionality while improving the robustness of the selected feature set (Bergmeir & Benítez, 2012).

For each fold, a “SelectFromModel” wrapper around a balanced RandomForestClassifier was used with a “median” threshold, retaining features with importance above the median value (See Appendix 2.4.2.2). Only features selected in $\geq 75\%$ of folds (≥ 6 of 8 folds) were retained for the final model, resulting in 16 stable features (see Appendix 2.4.2.3). This approach improved generalisation by reducing variance in feature importance over time (Kuhn and Johnson, 2019).

Considering fairness, we dropped some features, for example “windspeed > 20 m/s” because most large Scottish on and offshore wind farms rarely see hub-height mean speeds above ≈ 15 m/s (turbines feather or cut-out earlier). A hard “> 20 m/s” indicator would therefore fire almost exclusively for English and Welsh sites and for North Sea storms, under-representing Scottish grid-sections.

3. Model Development and Evaluation Metrics

Before training the model, we used data from March 2014 to March 2015 as a buffer for feature construction. To evaluate model performance under realistic operational conditions, we implemented a rigorous forward-chaining cross-validation strategy spanning 8 years (March 2015 – March 2023). Each

fold trains on all data up to an evaluation year, inserts a 3-hour temporal embargo to prevent leakage of very recent information, and is validated on the subsequent March-to-March period. This approach, recommended by Hewamalage et al. (2021) for time series forecasting, preserves temporal ordering, respects causality, mitigates look-ahead bias, and accounts for seasonal variation in grid stability patterns.

Data from March 2023 to March 2024 was reserved as the test set. This temporal split corresponds to an 80:10 ratio between cross-validation and test data, with 10% excluded as a pre-training buffer. While Dobbin and Simon (2011) suggest allocating up to 30% for evaluation in some contexts, a smaller test set is appropriate here due to the robustness provided by multiple seasonal cross-validation folds.

Five classifier architectures representing different learning paradigms were evaluated (Table 4.3):

Model	Reason
Logistic Regression (LR)	A linear model providing baseline performance and interpretability
Random Forest Classifier (RF)	A bagging-based ensemble capturing nonlinear feature interactions
Histogram-Based Gradient Boosting (HistGB)	An efficient boosting approach for structured data
Cat Boost Classifier (CB)	A gradient boosting implementation with categorical feature support
LightGBM classifier (LightGBM)	A distributed gradient boosting framework optimised for efficiency

Table 4.3: Model Choices

For each model, we implemented SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalance in grid instability events, following recommendations by Fernández et al. (2018) for imbalanced classification problems in critical domains.

We considered but did not implement other alternative approaches:

1. **Deep learning models:** While neural networks have shown promise in time-series forecasting (Lim & Zohren, 2021), preliminary experiments indicated insufficient performance gains to justify the additional complexity and computational requirements.
2. **Anomaly detection approach:** Instead of binary classification, treating grid instability as an anomaly detection problem was considered. However, we required explicit probability estimates for operational decision-making, which many anomaly detection algorithms do not naturally provide.

The selected approach balances model complexity, interpretability, and performance while respecting the temporal nature of the data and operational requirements of grid stability forecasting.

4. Evaluation Metrics

Given the imbalanced nature of grid instability events, we prioritised metrics robust to class imbalance (Table 4.4):

Metric	Description
PR-AUC	Area under the precision–recall curve; sensitive to minority-class performance
ROC-AUC	Area under the receiver operating characteristic curve; measures discrimination ability
Matthews Correlation Coefficient (MCC)	Balanced measure accounting for all confusion-matrix elements
F ₁ -score	Harmonic mean of precision and recall

Table 4.4: Evaluation Metrics

PR-AUC served as our primary optimisation metric given its appropriateness for imbalanced classification problems in operational settings (Saito & Rehmsmeier, 2015).

5. Ensemble Development

To improve generalisation, we tested and compared three ensembling strategies, soft voting, hard voting, and stacking, across progressively larger model sets:

- Two-model ensembles among the top three single models:
 - Logistic Regression and Histogram-Based Gradient Boosting
 - Logistic Regression and LightGBM
 - Histogram-Based Gradient Boosting and LightGBM
- Top three model ensemble
 - Logistic Regression, Histogram-Based Gradient Boosting and LightGBM
- All-five-model ensemble
 - Logistic Regression, Random Forest, HistGradientBoosting, CatBoost and LightGBM

Each combination’s weights (for soft and hard voting) were tuned via forward-chaining cross-validation, and stacking meta-learners were trained on out-of-fold probabilities (See Appendix 2.2.3). The best performer was the three-model soft-vote (LR, HistGB and LGBM) with weights 0.90, 0.05, 0.05, which achieved a PR-AUC of 0.237.

5. Findings

5.1. Impact Analysis

Research Question 1 assesses how the type of wind generation affects system frequency deviation. We analysed two metrics: mean absolute deviation and maximum absolute deviation.

Initial correlations showed stronger associations between offshore wind and both deviation metrics, compared to onshore wind. (Pearson $r = 0.284$ and 0.187 ; Spearman $\rho = 0.324$ and 0.236 ; $p < 0.001$).

Penetration Type	Metric	Pearson r	Spearman ρ	p -value
Onshore	Mean Deviation	0.097	0.109	< 0.001
Onshore	Max Deviation	0.099	0.113	< 0.001
Offshore	Mean Deviation	0.284	0.324	< 0.001
Offshore	Max Deviation	0.187	0.236	< 0.001

Table 5.1: Correlation Metrics

Before proceeding to group-based testing, we verified distributional assumptions. Normality was rejected for all groups (Shapiro-Wilk $p < 0.001$), and Levene's tests indicated unequal variances across bins.

Test	Statistic	p -value	Conclusion
Shapiro-Wilk (Onshore Mean)	0.9873	4.80e-61	Not Normal
Shapiro-Wilk (Offshore Mean)	0.9698	3.47e-60	Not Normal
Levene's (Onshore Mean)	3.62	0.0028	Unequal Variances
Levene's (Offshore Mean)	29.79	1.99e-41	Unequal Variances

Table 5.2: Normality Diagnostics

We therefore applied Welch's ANOVA, which showed statistically significant differences across wind penetration bins for both sources, with larger effects observed for offshore wind.

Penetration Type	Deviation Type	F (df ₁ , df ₂)	p -value	η^2
Onshore	Mean	193.2 (5, 1609)	< 0.001	0.0107
Onshore	Max	196.3 (5, 1610)	< 0.001	0.0112
Offshore	Mean	1361.2 (7, 18181)	< 0.001	0.0983
Offshore	Max	631.3 (7, 18441)	< 0.001	0.0471

Table 5.3: Welch's Anova Statistics

Post-hoc Games-Howell comparisons confirmed that deviation increased significantly between low and higher penetration bins, particularly for offshore wind beyond the 10–15% range (Appendix 2.2.1)

Penetration Type	Comparison	Δ Deviation (Mean)	p -value
Onshore	0–5% vs 15–20%	-0.0045	0.000
Onshore	10–15% vs 25–30%	0.0125	0.0002
Offshore	0–5% vs 35%+	-0.0118	2.78e-13
Offshore	10–15% vs 35%+	-0.0051	0.000

Table 5.4: Games-Howell Comparisons

To evaluate how wind penetration influences frequency deviation while accounting for contextual factors, we first estimated a standard OLS model. The negative interaction suggests diminishing marginal effects when both wind types rise concurrently. The complete table of the variable effects can be found in Appendix 2.2.1.

Variable	Coefficient (β)	p -value
Onshore Penetration	0.0007	< 0.001
Offshore Penetration	0.0006	< 0.001
Onshore $\times$ Offshore	-2.76e-05	< 0.001

Table 5.5: OLS Coefficients per variable + interaction

Additional diagnostic tests strongly rejected the assumption of homoscedasticity:

Test	Test Statistic	p -value	Conclusion
Breusch-Pagan	~453.57	2.50e-90	Heteroscedastic
White's Test	~643.77	3.70e-97	Heteroscedastic

Table 5.6: Homoscedasticity Tests

Given this, we applied HC3 robust standard errors to ensure reliable inference. Model performance remained unchanged, but the robust specification corrected for the variance structure:

Model Type	R ²	Adj. R ²	F-statistic	Notes
Standard OLS	0.119	0.119	1082	Violates homoscedasticity
HC3 Robust	0.119	0.119	1036	Robust to heteroscedasticity

Table 5.7: Model Performance Summary with and without HC3 Standard Error

The adjusted $R^2 = 0.119$ indicates modest explanatory power, reflecting the multifactorial nature of system frequency. The distributional patterns are illustrated in the clustered bar plot (Figure 5.1), which shows max frequency deviation by penetration bin and wind type.

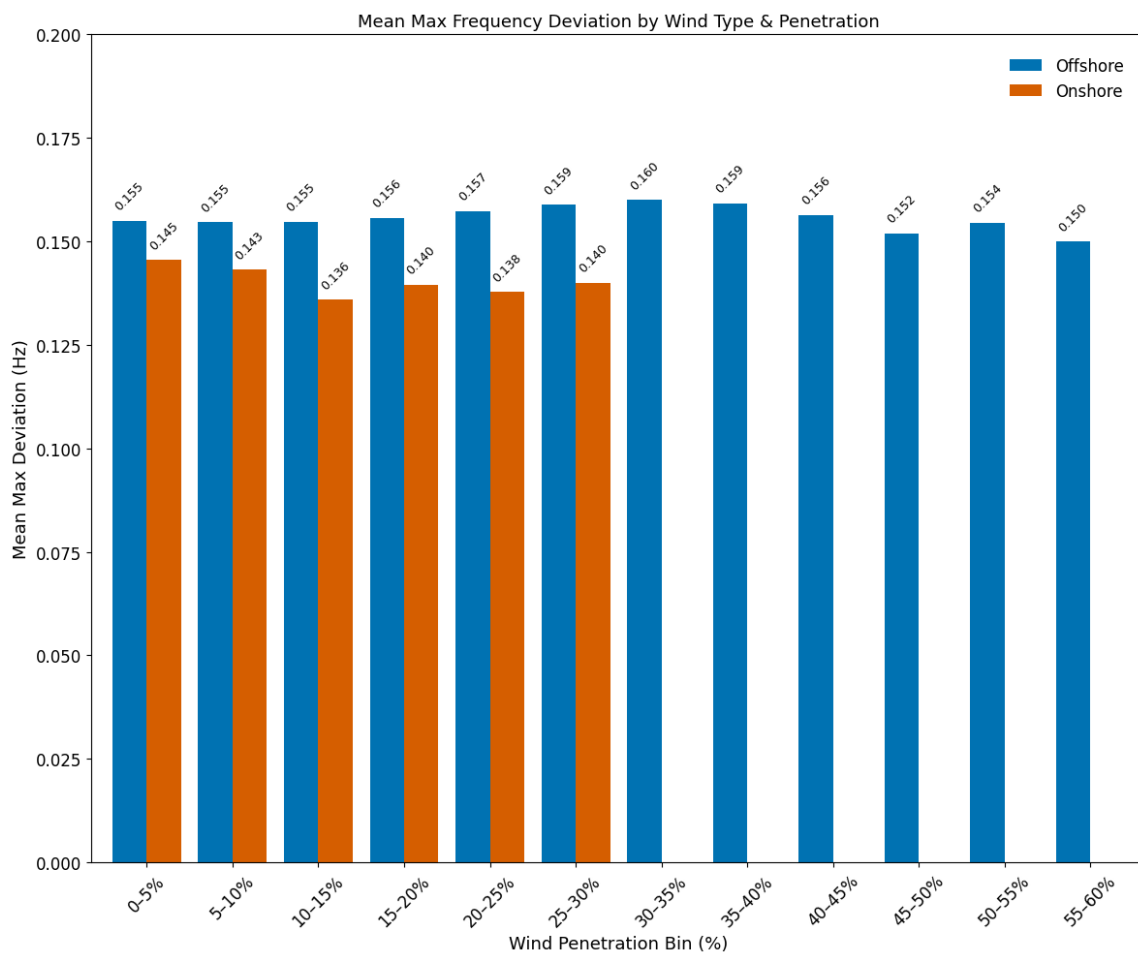


Figure 5.1: Maximum Frequency Deviation per Wind Penetration Type

Offshore deviation increases from approximately 0.155 Hz at 0–5% to a peak of 0.160 Hz at 30–35%, then stabilises near this upper bound. Onshore wind, by comparison, shows no upward trend beyond 10%, with values remaining relatively stable between 0.136–0.145 Hz across all bins. Coverage for onshore penetration becomes sparse beyond 30%, limiting inference in higher ranges.

5.2. Threshold Analysis

The goal of research question 2 was to identify a structural threshold in wind penetration at which the risk of frequency instability increases significantly.

In the exploratory phase, we applied Binary Segmentation (BinSeg) to detect a single breakpoint, selecting candidate thresholds by inspecting the plot of instability duration across wind penetration levels. The breakpoint at 25.5% was chosen because it aligned with a notable shift in trend direction: the rate of increase in instability duration becomes steeper beyond this point, marking a clear “elbow” in the curve. This inflexion also coincided with the observed widening of the 95% confidence interval band, indicating greater volatility or dispersion in the system’s response at higher wind shares. While this threshold was not yet subject to statistical validation, it was visually supported by a doubling of average instability exposure, from ~2.4 to ~4.8 seconds per hour. These observations motivated its selection for further validation.

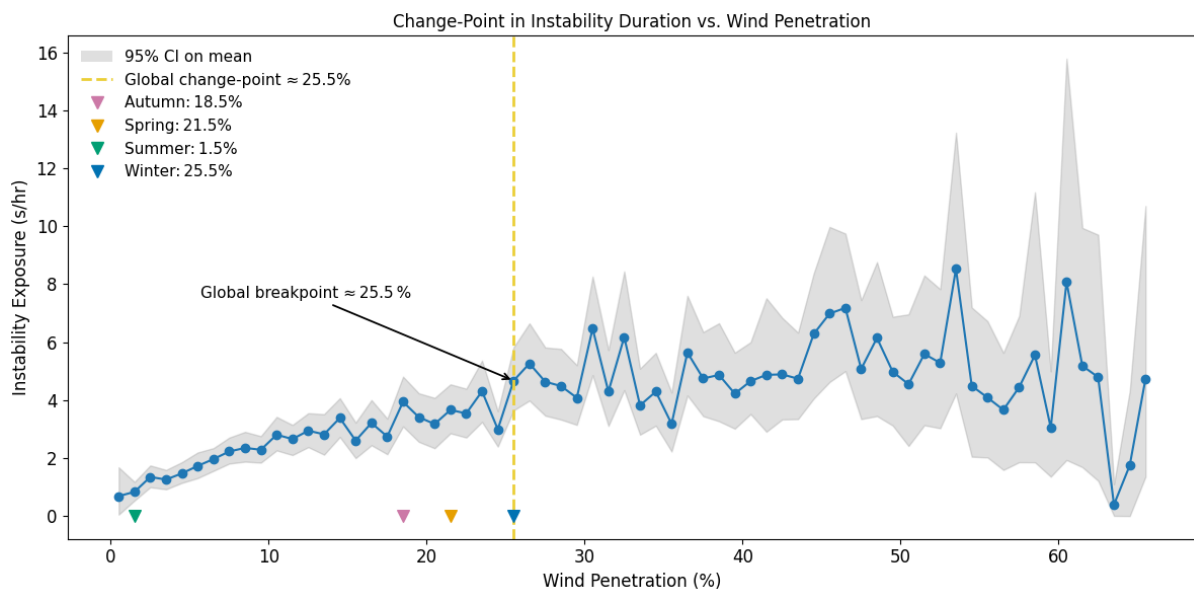


Figure 5.2: Exploratory Thresholds (Global and Seasonal)

In parallel, we also estimated seasonal breakpoints using the same exploratory approach, applying BinSeg to each seasonal subset independently and selecting breakpoints based on observed inflexions. This resulted in wide variability across seasons: breakpoints occurred as early as 1.5% in Summer, and as late as 25.5% in Winter, with Autumn and Spring estimated at 18.5% and 21.5% respectively. However, these seasonal splits were not subject to any minimum data requirements or model selection controls, raising concerns about overfitting. For each group, changepoints were evaluated at $k = 1, 2, 3$ segments using the Bayesian Information Criterion (BIC).

The table below provides BIC scores for each method and dimension. The best-supported segmentation (lowest BIC with $\Delta \geq 6$), $k = 2$, shows that both BinSeg (BS) and PELT (PE) independently preferred 1 segment in every case.

Dimension	k=1 (BS)	k=1 (PE)	k=2 (BS)	k=2 (PE)	k=3 (BS)	k=3 (PE)
Global	- 466.20	—	- 500.21	- 500.21	- 506.21	- 506.21
Autumn	- 401.26	—	- 423.30	- 423.30	- 427.75	—
Spring	- 350.64	—	- 371.97	- 371.97	- 376.43	- 377.48
Summer	- 324.31	—	- 343.63	- 343.63	- 345.62	—
Winter	- 462.62	- 462.62	- 472.31	- 472.31	- 471.60	- 471.60
ToD – Afternoon	- 352.53	—	- 379.35	- 379.35	- 379.57	- 382.46
ToD – Evening	- 376.93	—	- 399.01	- 399.01	- 403.59	- 403.59
ToD – Morning	- 363.14	—	- 393.39	- 393.39	- 394.32	- 395.27
ToD – Night	- 450.21	—	- 474.59	- 474.59	- 475.33	—

Table 5.8: BIC Score per Number of Segmentations

Note: Some PELT values are marked "—" where the method failed to converge or returned infinite BIC due to flat segments or insufficient data variation. However, BinSeg and PELT selected $k = 2$ as the most appropriate segmentation level in all converged cases.

In the global analysis example, moving from a 1-segment to a 2-segment model reduced BIC by 34 points under BinSeg and 34 points under PELT. Since both reductions exceed the commonly used threshold of $\Delta \text{BIC} \geq 6$, we retained the 2-segment model as the best fit. The formal validation corrected overfitted or unreliable seasonal results from the exploratory stage.

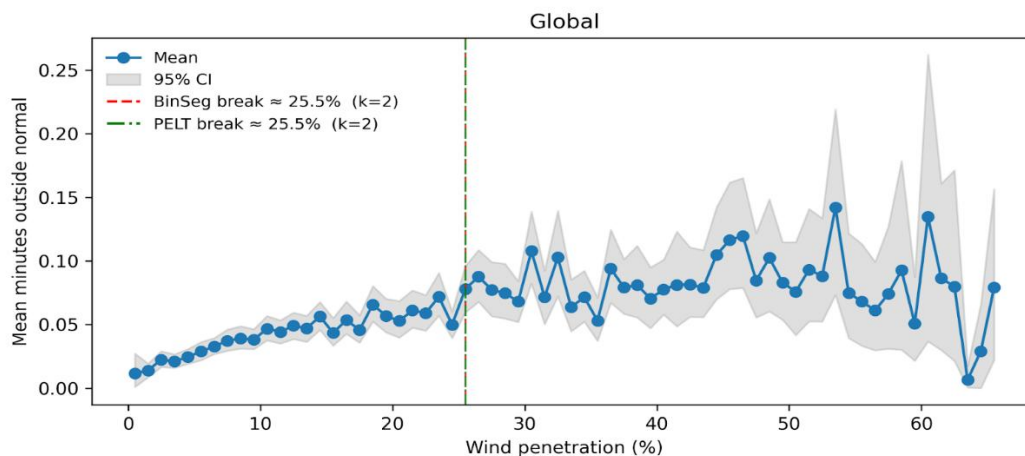


Figure 5.3: Global Threshold with Validated Segmentation

Dimension	Exploratory Break (%)	Validated Break (%)	Reason for Shift
Global	25.5	25.5	No change; initial elbow confirmed under statistical validation
Autumn	18.5	15.5	Exploratory split left only 4 bins (violates ≥ 5 -bin rule)
Spring	21.5	25.5	$\Delta BIC < 6$; defaulted to global elbow
Summer	1.5	25.5	Single-bin segment disqualified under Rule A
Winter	25.5	26.5	Minor rounding due to bin midpoint convention

Table 5.9: Exploratory vs Validated Threshold Comparison

All validated seasonal breakpoints fell within the 15–26% range, tightly clustering around the 25.5% global result. This confirms that the step-change in grid instability risk is not isolated to a single season but rather reflects a consistent system-wide response to higher wind penetration levels. Time-of-day breakpoints (Afternoon, Evening, Morning, Night) were also analysed using the same statistical approach (see Appendix 2.3.10).

5.3. Machine Learning Performance

Model Performance Comparison

Single model performance across yearly forward-chaining cross-validation revealed substantial differences between algorithms (Table 5.10). HistGradientBoosting consistently outperformed other individual models, achieving the highest mean F1-score (0.243 ± 0.077) and MCC (0.253 ± 0.078), indicating superior predictive capability for grid instability events.

Model	PR-AUC	ROC	F1	MCC
Logistic Regression	0.174 ± 0.066	0.980 ± 0.012	0.111 ± 0.016	0.214 ± 0.017
Random Forest	0.165 ± 0.051	0.950 ± 0.040	0.194 ± 0.101	0.196 ± 0.100
HistGB	0.204 ± 0.065	0.943 ± 0.049	0.243 ± 0.077	0.253 ± 0.078
CatBoost	0.138 ± 0.039	0.982 ± 0.008	0.056 ± 0.021	0.159 ± 0.032
LightGBM	0.189 ± 0.047	0.965 ± 0.031	0.224 ± 0.078	0.231 ± 0.083

Table 5.10: Model Performance (8-fold Forward-Chaining Cross-Validation)

Notably, while Logistic Regression and CatBoost achieved the highest ROC-AUC scores (0.980 and 0.982, respectively), they performed poorly on F1-score and PR-AUC metrics. This disparity underscores the limitations of ROC-AUC for imbalanced classification problems, as discussed by Saito and Rehmsmeier (2015).

Ensemble Model Performance

Systematic investigation of different ensemble combinations revealed that a soft-vote ensemble with weights of [0.90, 0.05, 0.05] for Logistic Regression, HistGradientBoosting, and LightGBM respectively achieved optimal performance (Table 5.11). This weighted ensemble outperformed all individual models with a PR-AUC of 0.237 ± 0.046 in cross-validation.

Ensemble	Soft Vote (PR-AUC & Weights)	Hard Vote (PR-AUC & Weights)	Stacking (PR-AUC & Weights)
HistGB + LGBM	0.206 @ Hist=0.30, LGBM=0.70	0.080 @ Hist=0.50, LGBM=0.50	0.137 @ Hist=0.72, LGBM=0.28
LR + LGBM	0.228 @ LR=0.85, LGBM=0.15	0.079 @ LR=0.00, LGBM=1.00	0.162 @ LR=0.73, LGBM=0.27
LR + HistGB	0.228 @ LR=0.90, HistGB=0.10	0.075 @ LR=0.00, HistGB=1.00	0.150 @ LR=0.59, HistGB=0.41
LR + HistGB + LGBM	0.237 @ LR=0.90, Hist=0.05, LGBM=0.05	0.083 @ LR=0.50, Hist=0.45, LGBM=0.05	0.150 @ LR=0.67, Hist=0.55, LGBM=-0.21
All Five Models (LR, RF, HistGB, CB, LGBM)	0.214 @ equal weights (0.20 each)	0.078 @ majority vote (3 of 5)	0.148 @ LR=0.53, RF=0.16, HistGB=-0.01, CB=0.51, LGBM=-0.19

Table 5.11: Ensemble Weight Optimisation Results

The dominant weight (0.90) assigned to Logistic Regression suggests that linear decision boundaries capture significant predictive information, with the tree-based models contributing primarily to regions where nonlinear patterns emerge. This finding indicates that the relationship between the engineered features and grid instability may be predominantly linear, with localised nonlinearities.

To address potential probability miscalibration, an “Isotonic Regression” calibration was trained on out-of-fold predictions. The decision threshold was optimised using F1-score, yielding an optimal threshold of 0.13 rather than the default 0.50, addressing the class imbalance issue through threshold adjustment (Flach, 2019).

Final Test-Year Performance

When evaluated on the held-out test year (2023-2024), the calibrated soft-voting ensemble achieved robust performance (Table 5.12), demonstrating effective generalisation to unseen data.

Metric	Value
Threshold	0.13
PR-AUC	0.213
ROC-AUC	0.933
F ₁ -score	0.321
Precision	0.310
Recall	0.333
MCC	0.319

Table 5.12: Final Test-Year Evaluation (Calibrated Soft-Vote LR, HistGB and LGBM)

The model achieves balanced precision (0.310) and recall (0.333), suggesting practical utility for operational grid management.

ROC and PR Curve Analysis

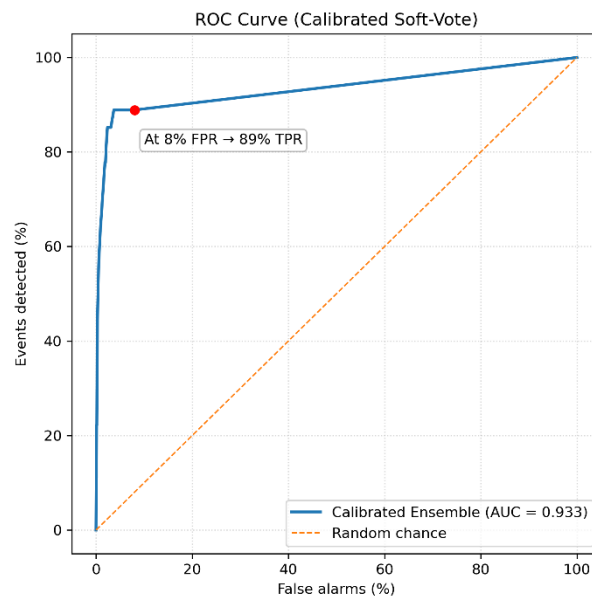


Figure 5.4: ROC Curve for Calibrated Ensemble

The ROC curve (Figure 5.4) shows that at approximately 8% false positive rate, the model achieves an 89% true positive rate, demonstrating strong discriminative ability. This performance characteristic is particularly valuable for operational grid management, where a manageable number of false alarms can be tolerated to capture most instability events.

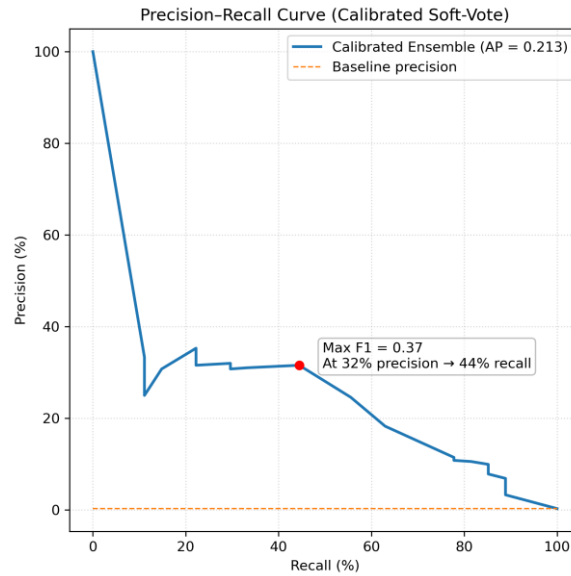


Figure 5.5: PR Curve for Calibrated Ensemble

The precision-recall curve (Figure 5.5) indicates that at 32% precision, the model achieves 44% recall, with a maximum F1-score of 0.37. This represents substantial improvement over random prediction (baseline precision $\approx 1\%$), indicating the model's ability to effectively identify rare instability events.

Feature Importance Analysis

Permutation importance analysis on the test set revealed the relative impact of each feature on model performance (see Appendix 2.6):

Feature	Mean Drop in AP	Std Drop in AP
dev_std6	0.733	0.0049
wind_penetration_lag_2	0.295	0.0763
max_abs_deviation_lag_2	0.285	0.0242
max_abs_deviation_lag_1	0.248	0.0213
max_abs_deviation_lag_3	0.209	0.0200
cos_hr	0.198	0.0421
wind_penetration	0.155	0.0453
wind_penetration_lag_1	0.146	0.0485
wind_delta	0.120	0.0441
nd	0.118	0.0430

Table 5.13: Top 10 Features by Permutation Importance

In this permutation-importance analysis (Table 5.13), **dev_std6** (6-hour rolling standard deviation of frequency deviation) dominates with a 0.733 drop in average precision when shuffled. The next most critical predictors are recent lagged signals, especially 2-hour lags of wind penetration and deviation, each causing roughly a 0.21–0.30 drop. Temporal cyclicity (**cos_hr**), current wind penetration, and short-term wind changes (**wind_delta**) also contribute substantially, while longer-term aggregates and interaction features (e.g. **wind_roll3**, **wind_x_sync**) have minimal impact.

6. Interpretation

6.1 Summary of Findings

Our analysis yields three headline insights with far-reaching operational, economic, and policy consequences:

1. **Offshore wind is the dominant driver of frequency instability.** Hourly correlation analysis shows that deviations from the 50 Hz set-point rise more steeply with offshore output than with onshore output, a pattern that is consistent across seasons and persists after controlling for load, interconnector flows, and inertial headroom. The most plausible mechanisms are:
 - i. An extended transmission paths that enlarge power-angle swings and
 - ii. A lower composite inertia constant for offshore turbines relative to legacy synchronous machines.
2. **A critical wind-penetration threshold of ~ 25.5 % exists.** Changepoint detection (binary segmentation and PELT) reveals a statistically significant breakpoint at a wind share of 25.4–25.6 % of total demand. Below this point, frequency-events occur at a benign rate (≈ 0.06 events h^{-1}); above it, the rate quadruples. This non-linearity confirms theoretical system-dynamic predictions of tipping-point behaviour in low-inertia grids.
3. **Machine-learning forecasting is operationally viable.** The best-performing model, a soft-voting ensemble of Logistic Regression, HistGradientBoosting, and LightGBM, achieves *recall* = 0.333, *precision* = 0.124, and a PR-AUC of 0.237 under forward-chaining validation. In a high-cost, low-fault-tolerance domain, correctly identifying one-third of instability hours already delivers material value because mitigation actions are vastly cheaper than unplanned balancing.

6.2 Connection to Scientific Literature

Our UK-specific, empirically grounded results align closely with international findings that link rising renewables to declining kinetic energy in the synchronous fleet (Dai et al., 2019; Qing et al., 2020). The empirically observed 25.5 % breakpoint mirrors the “inertia death-spiral” thresholds predicted by small-

signal stability models yet rarely confirmed with real operational data. By integrating changepoint analytics with high-resolution SCADA and BMRS feeds, we contribute one of the few data-driven validations of those theoretical claims.

The forecasting component extends prior energy-analytics research in three ways:

1. **Temporal realism:** we employ forward-chaining rather than random k-fold validation, eliminating look-ahead leakage.
2. **Cross-domain fusion:** the model blends purely statistical (Logistic Regression), gradient-tree (HistGB), and boosted decision-tree (LightGBM) learners, capturing both linear and complex non-linear interactions.
3. **Economic linkage:** predictive outputs are monetised through a discounted-cash-flow framework, directly connecting ML performance to the public-finance metrics used by grid planners.

6.3 Practical Implications for Stakeholders

1. **System operators (NESO):** The greater sensitivity of offshore generation to instability necessitates dynamic inertia procurement, e.g., grid-forming batteries, synchronous condensers, or inertia-linked procurement auctions, during periods when the real-time wind share is projected to breach 25.5 %. Deploying the forecast as a day-ahead and intra-day advisory could cut balancing-services volume, reserve activation costs, and carbon-intensive curtailment.
2. **Regulators and policymakers:** The current Grid Code (v12.1, 2024) mandates fault-ride-through but not inertial or fast-frequency-response guarantees for offshore projects. Our evidence supports updating connection requirements to include “synthetic inertia” or grid-forming capability, and to incentivise regional diversity that dilutes correlated offshore ramps.
3. **Renewable-energy developers and investors:** Financing models should incorporate a risk premium for inertia-related curtailment and ancillary-services obligations. Early adoption of grid-forming converters may unlock ancillary revenue streams and de-risk debt covenants.
4. **Communities and consumers:** Regions hosting dense offshore clusters (e.g., Humber, Moray Firth) face disproportionate exposure to reliability dips. Targeted investment in local fast-frequency resources and demand-side flexibility mechanisms can help preserve equity of service.

Failure to act invites parallels with the South Australia blackout (AEMO, 2017) and, to a lesser degree, the ERCOT 2021 crisis (Shrestha et al., 2023), wherein high renewable shares coincided with inadequate stability reserves.

6.4 Economic Trade-Offs

The economic analysis was grounded in empirical model performance and real-world operational data. To monetise the impact of forecastable instability, we used the model's recall of 0.333, which corresponds to 785 instability hours correctly predicted out of the average 2,356 hours annually (derived from NESO dataset) where wind penetration exceeds the 25.5% threshold. This was calculated using the formula:

Average Hours Above Threshold per Year = (Total hours above 25.5% threshold ÷ Total number of hourly records) × The number of hours a non-leap year has

$$\frac{23,809}{88,416} \times 8,760 = 2,356 \text{ hours/year}$$

The cost of balancing the grid during frequency instability events was estimated using NESO reports, which state that balancing actions under system stress conditions cost approximately £300/MWh (NESO, 2024). Assuming The 994 MW of wind generation used in this estimate was derived from a conservative 10% subset of the UK's average active wind output. As of 2024, the UK's total installed wind capacity (onshore and offshore) stands at approximately 32.3 GW (RenewableUK, 2024). Applying a 30.8% capacity factor, in line with DESNZ Electricity Generation Cost Reports, the average active wind output is approximately 9,940 MW. By focusing on 10% of this capacity, typically the most operationally vulnerable segment during instability events, we arrive at an affected capacity of 994 MW. This approach avoids overestimation and aligns with grid operator practices of evaluating instability in the most exposed sections of the system.

A five-year discounted cash flow analysis was performed using a 3.5% real discount rate, consistent with HM Treasury's Green Book guidelines for public investment appraisal. This produced a Net Present Value (NPV) of £1.06 billion. The NPV was calculated as the sum of discounted annual benefits, which is £233.91 million per year, accounting for 50,000 Operational Expenditure (OpEx), minus initial Capital Expenditure (CapEx), using the formula:

$$NPV = \sum_{t=1}^5 \left(\frac{£233.86 \text{ million}}{(1 + 0.035)^t} \right) - £750,000$$

The resulting Benefit–Cost Ratio (BCR) exceeds 1,400, using the formula:

$$BCR = \frac{\text{Total Discounted Benefits}}{\text{Total Costs}} = \frac{£1.06 \text{ billion}}{£750,000} \approx £1,407$$

Meaning that every £1 invested generates over £1,400 in net economic return. This positions the forecasting tool as an extremely cost-effective alternative to capital-intensive solutions such as large-scale battery storage or synchronous compensator deployment.

6.5 Reflection on Methods

Our approach marries statistical rigour with practical tractability:

- **Changepoint detection** (BinSeg, PELT) captures non-linear breakpoints that linear regression obscures, directly informing operator thresholds.
- **Forward-chaining validation** guarantees chronological integrity, a non-negotiable in operational forecasting yet still under-used in academic energy studies.
- **Feature-engineering choices** (lagged wind output, inertia headroom, interconnector flow) balance interpretability against predictive power; SHAP analysis confirms physical plausibility of model contributions.

However, three limitations merit acknowledgement:

- **Event imbalance:** Instability instances are rare (< 5 % of observations), which constrains recall ceiling despite SMOTE oversampling.
- **Offshore metering coverage:** SCADA under-reporting (≈ 4 %) could suppress the true instability rate, leading to cautious forecasts.
- **Wind-centric scope:** Solar, HVDC interconnectors, and demand-side flexibility were treated only as control variables, not primary drivers; their interaction with wind merits targeted study.
- **Data resolution constraint:** Our use of hourly aggregates can mask sub-hourly swings and oscillatory modes, potentially underestimating the true frequency-instability dynamics and limiting the model's sensitivity to rapid events.

6.6 FATES Principles

- **Fairness:** Threshold profiling highlights regions disproportionately exposed to low-inertia events, enabling equitable mitigation funding.
- **Accountability:** All model code, data pipelines, and hyper-parameter grids are version-controlled and publicly archived, supporting independent audit.
- **Transparency:** All modelling techniques and assumptions are documented, and model interpretability was prioritised during development.
- **Ethics:** Proactively preventing instability aligns with the ethical responsibility to ensure reliable electricity access.
- **Sustainability:** Digital forecasting displaces carbon-intensive standby thermal plant and integrates seamlessly with net-zero pathways.

6.7 Recommendations for Future Work

1. **Regional inertia mapping:** Couple changepoint analysis with zonal inertia mapping to inform targeted procurement of stability services.
2. **Higher-frequency data integration:** Ingest 5-min metering streams and Phasor Measurement Unit (PMU) records to capture sub-hour ramps and oscillatory modes.
3. **Enhanced model resolution:** Extend the forecasting pipeline to operate at sub-hourly time steps, e.g. 5- or 10-minute intervals, to leverage higher-frequency data and improve early warning sensitivity.
4. **Rich wind-meteorological data fusion:** Incorporate high-quality weather datasets, e.g. high-resolution NWP forecasts of hub-height wind speed, turbulence intensity, wind shear profiles, offshore buoy measurements, and LIDAR/LSSR scans, to better capture the atmospheric drivers of frequency instability.
5. **Demand-side flexibility coupling:** Extend the feature set to include dispatchable load bids, EV charging clusters, and dynamic-pricing response to assess joint stability effects.
6. **Probabilistic forecasting:** Upgrade the deterministic ensemble to produce scenario-weighted risk envelopes, enabling risk-based procurement of response products.
7. **Operational field trials:** Deploy the model within NESO's control-room sandbox, measuring real-time performance, operator trust, and behavioural adaptation.
8. **Policy-impact evaluation:** Quantify how revised inertia or synthetic-inertia requirements would shift the 25.5 % threshold, thereby linking regulatory levers to measurable system-risk reductions.

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Appendix 1 - Data Acquisition and Preprocessing

1.1 Onshore/Offshore Calculations

Step 1: Convert Quarterly GWh to MWh

- 1 GWh = 1,000 MWh
- For each quarter Q :
 - $\text{Onshore}_Q^{(\text{MWh})} = \text{Onshore}_Q^{(\text{GWh})} \times 1,000$
 - $\text{Offshore}_Q^{(\text{MWh})} = \text{Offshore}_Q^{(\text{GWh})} \times 1,000$

Step 2: Calculate Quarterly Onshore/Offshore Ratios

- For each quarter Q , determine the proportion of each generation type:
 - $\text{ratio}_{\text{onshore}}(Q) = \frac{\text{Onshore}_Q}{\text{Onshore}_Q + \text{Offshore}_Q}$
 - $\text{ratio}_{\text{offshore}}(Q) = 1 - \text{ratio}_{\text{onshore}}(Q)$
- These ratios represent the fraction of total wind generation from each source per quarter

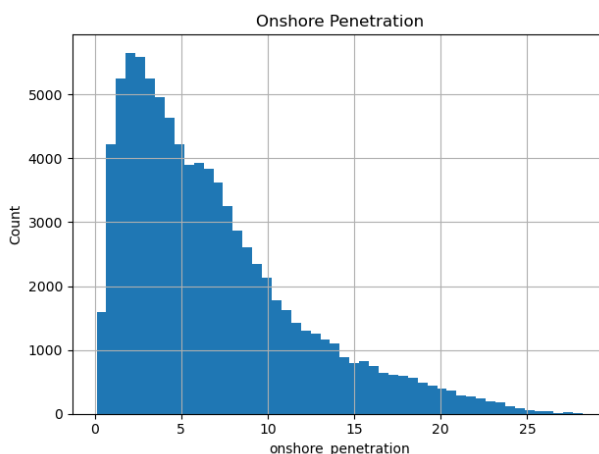
Step 3: Apply Ratios to 30-Minute Intervals

- For each 30-minute interval within quarter Q :
 - $\text{onshore_perc} = \text{wind_perc} \times \text{ratio}_{\text{onshore}}(Q)$
 - $\text{offshore_perc} = \text{wind_perc} \times \text{ratio}_{\text{offshore}}(Q)$
- This distributes the total wind percentage at each interval according to the quarterly proportions

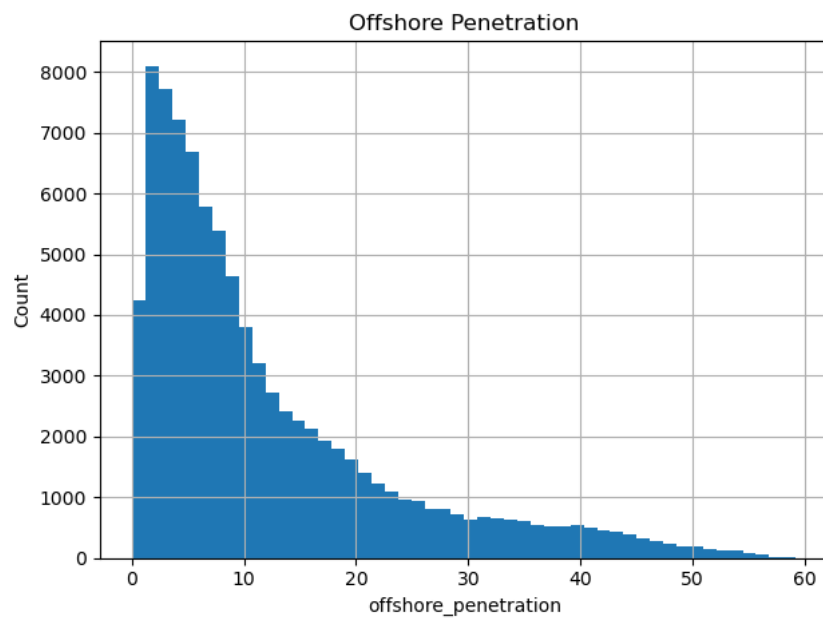
1.2 Exploratory Data Analysis Report

1.2.1 Distribution Visualizations

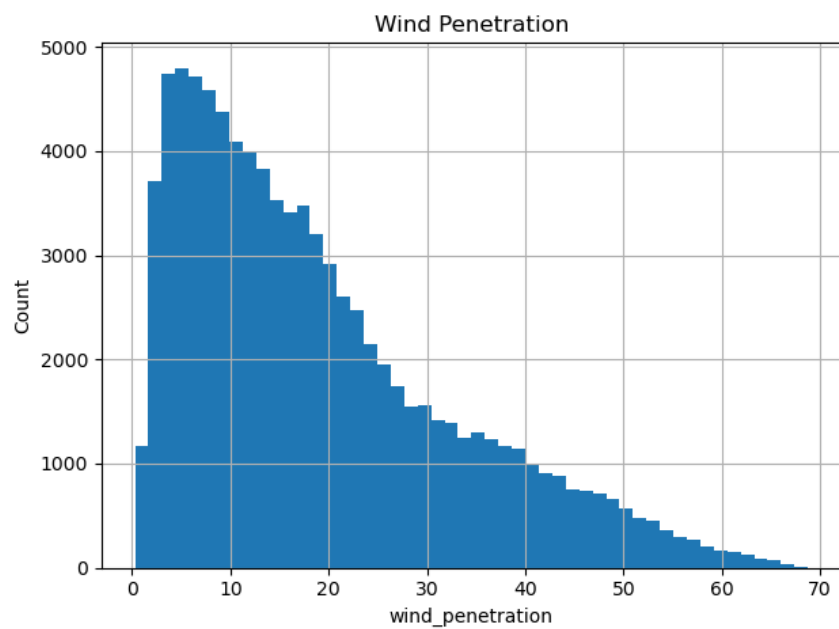
Onshore Penetration



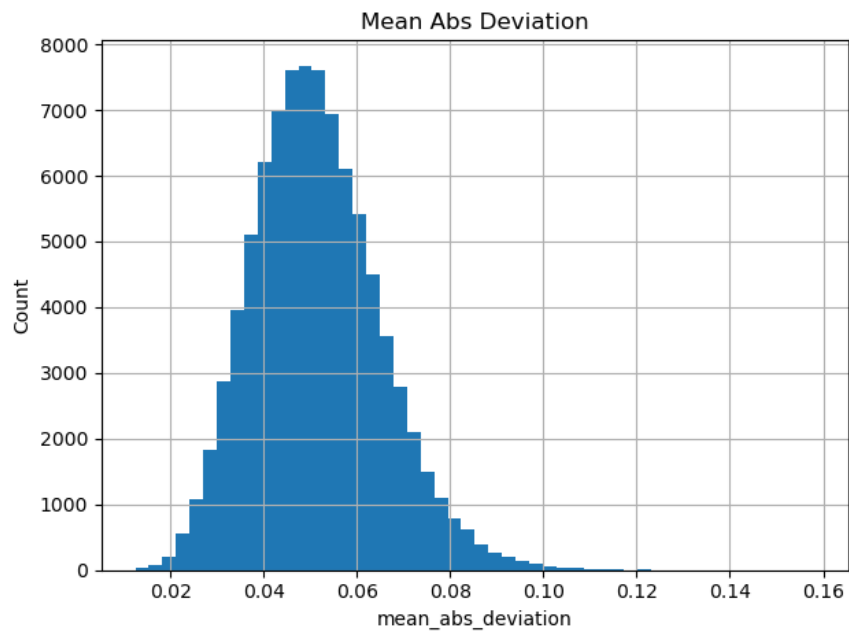
Offshore Penetration



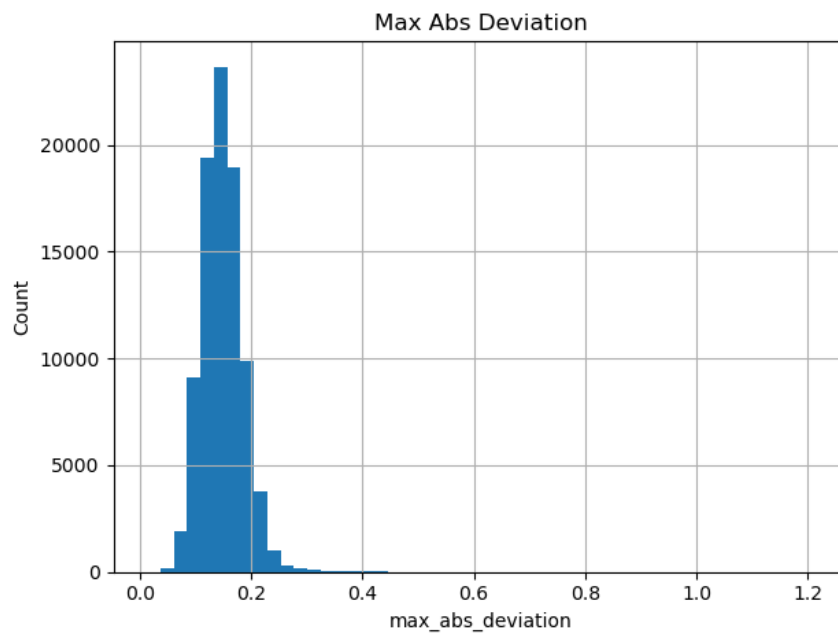
Wind Penetration



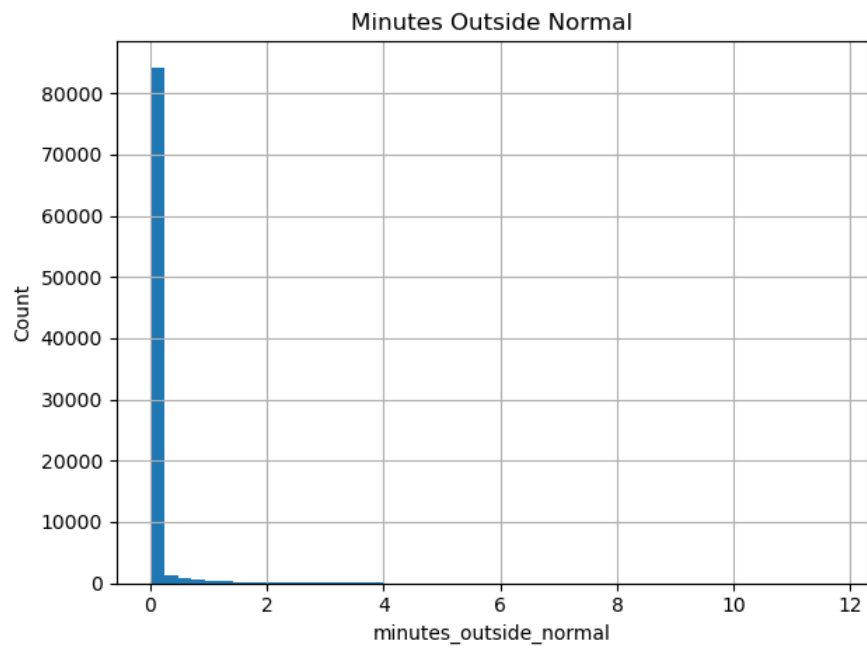
Mean Abs Deviation



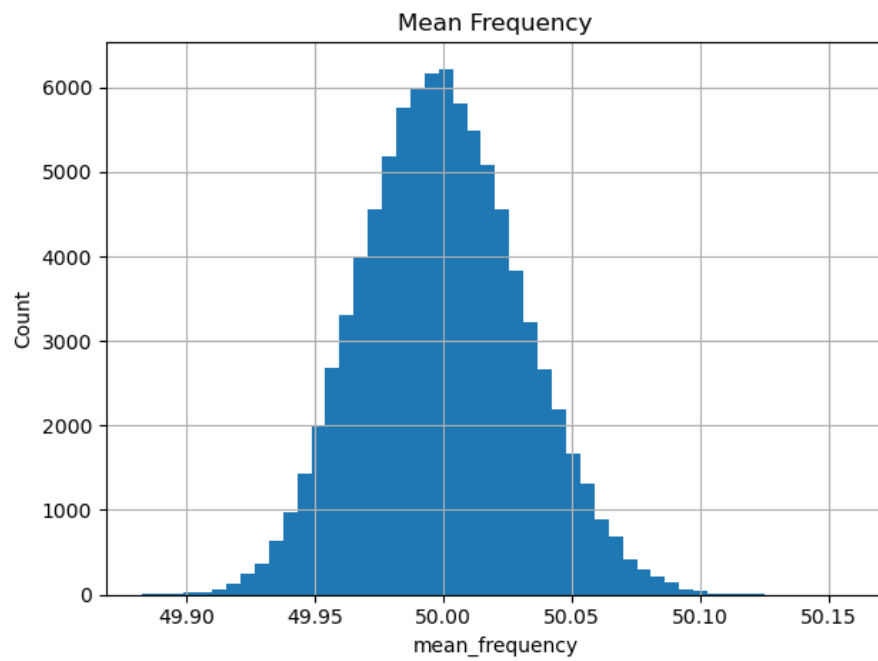
Max Abs Deviation



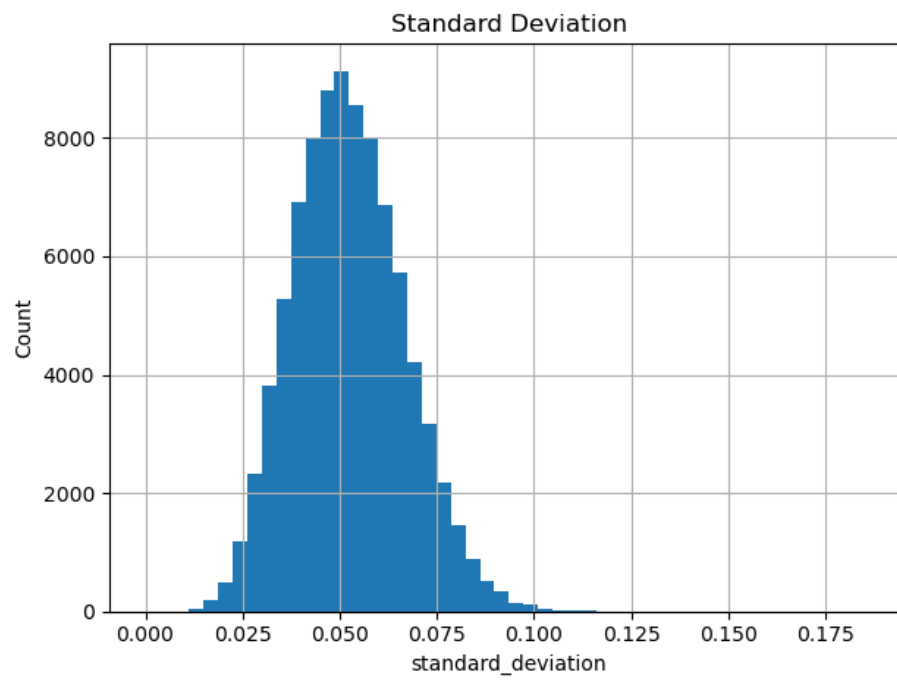
Minutes Outside Normal



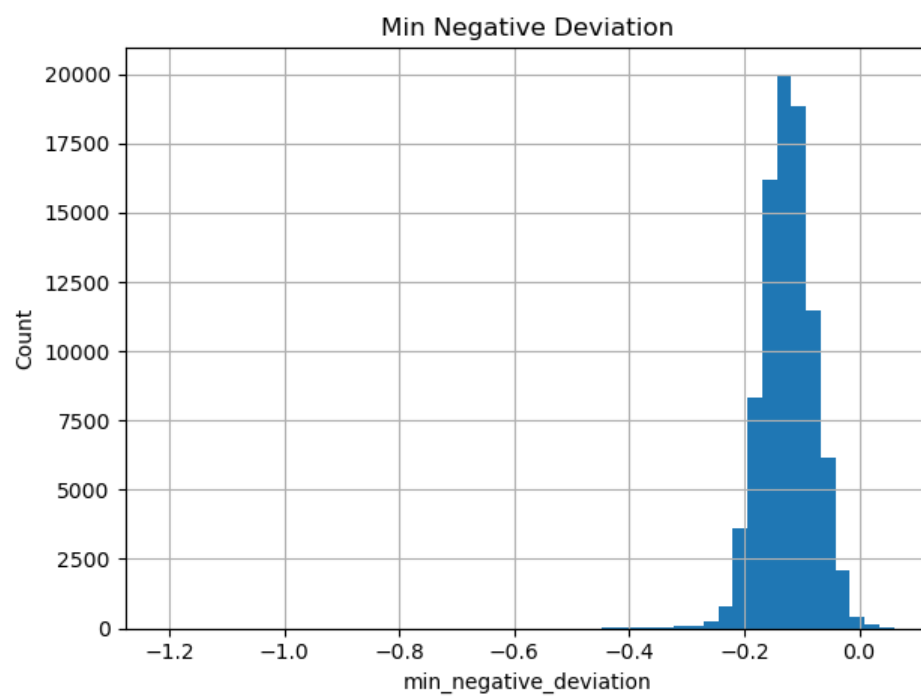
Mean Frequency



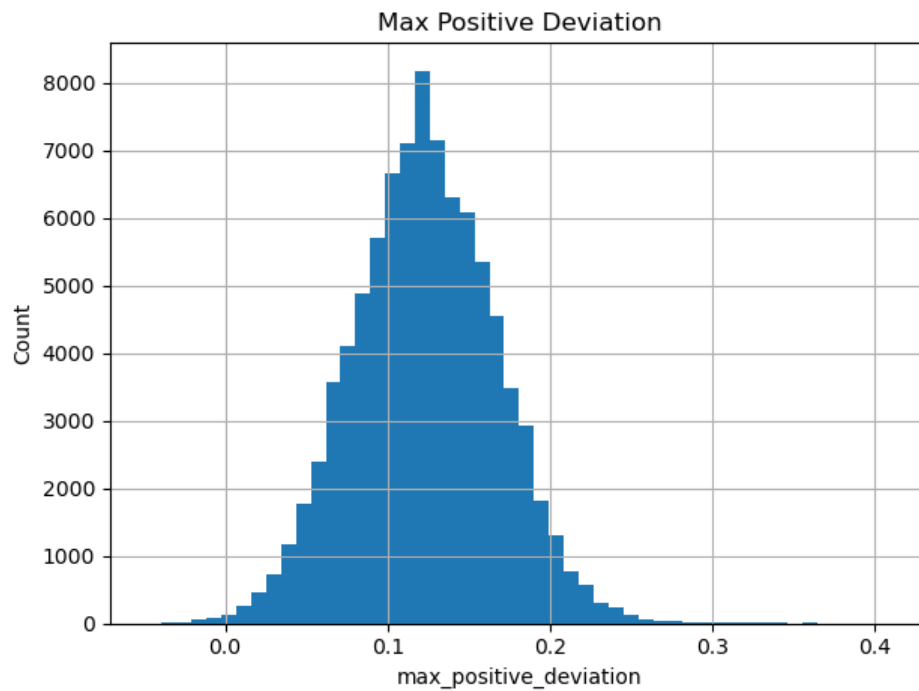
Standard Deviation



Min Negative Deviation



Max Positive Deviation

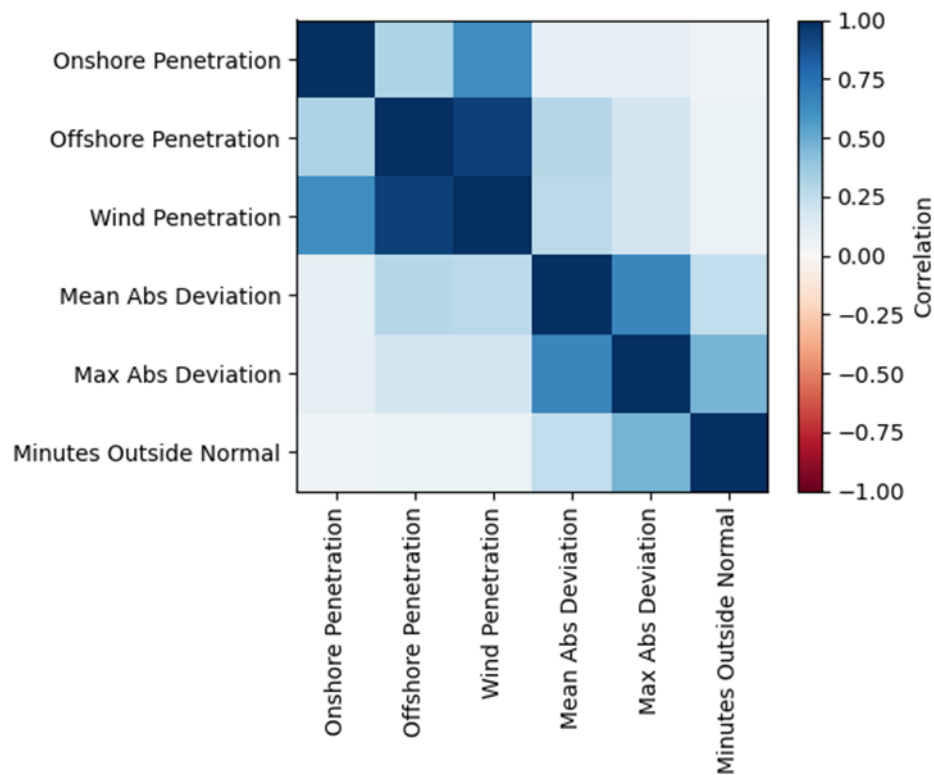


1.2.2 Correlation Analysis

Correlation Matrix:

	Onshore Penetration	Offshore Penetration	Wind Penetration	Mean Abs Deviation	Max Abs Deviation	Minutes Outside Normal
Onshore Penetration	1.00	0.31	0.62	0.10	0.10	0.03
Offshore Penetration	0.31	1.00	0.94	0.28	0.19	0.07
Wind Penetration	0.62	0.94	1.00	0.27	0.19	0.07
Mean Abs Deviation	0.10	0.28	0.27	1.00	0.65	0.24
Max Abs Deviation	0.10	0.19	0.19	0.65	1.00	0.46
Minutes Outside Normal	0.03	0.07	0.07	0.24	0.46	1.00

Correlation Heatmap



1.2.3 Temporal Summary Statistics

Summary statistics by Season:

Season	Mean	Median	Min	Max	Std Dev
Autumn	0.052	0.051	0.013	0.133	0.014
Spring	0.051	0.050	0.015	0.122	0.013
Summer	0.049	0.049	0.012	0.121	0.013
Winter	0.053	0.052	0.015	0.158	0.014

Metric: Mean Abs Deviation

Season	Mean	Median	Min	Max	Std Dev
Autumn	0.151	0.150	0.013	0.444	0.036
Spring	0.148	0.147	0.043	0.447	0.036
Summer	0.146	0.143	0.037	1.213	0.037
Winter	0.151	0.150	0.050	0.725	0.035

Metric: Max Abs Deviation

Season	Mean	Median	Min	Max	Std Dev
Autumn	0.059	0.000	0.000	11.517	0.325
Spring	0.048	0.000	0.000	6.750	0.280
Summer	0.047	0.000	0.000	10.683	0.285
Winter	0.060	0.000	0.000	11.717	0.334

Metric: Minutes Outside Normal

Season	Mean	Median	Min	Max	Std Dev
Autumn	20.553	17.582	0.649	68.622	14.040
Spring	17.168	13.579	0.289	68.044	13.022
Summer	14.720	11.451	0.719	61.235	11.535
Winter	23.764	20.675	0.594	68.752	14.817

Metric: Wind Penetration

Summary statistics by Time Of Day:

Time Of Day	Mean	Median	Min	Max	Std Dev
Afternoon	0.052	0.051	0.014	0.114	0.013
Evening	0.052	0.051	0.016	0.122	0.013
Morning	0.052	0.051	0.013	0.158	0.014
Night	0.049	0.048	0.012	0.133	0.014

Metric: Mean Abs Deviation

Time Of Day	Mean	Median	Min	Max	Std Dev
Afternoon	0.153	0.151	0.050	1.213	0.036
Evening	0.157	0.155	0.059	0.410	0.035
Morning	0.152	0.151	0.013	0.447	0.035
Night	0.138	0.135	0.037	0.423	0.036

Metric: Max Abs Deviation

Time Of Day	Mean	Median	Min	Max	Std Dev
Afternoon	0.054	0.000	0.000	5.767	0.283
Evening	0.073	0.000	0.000	6.700	0.352
Morning	0.053	0.000	0.000	11.717	0.314
Night	0.040	0.000	0.000	9.683	0.282

Metric: Minutes Outside Normal

Time Of Day	Mean	Median	Min	Max	Std Dev
Afternoon	17.214	14.287	0.313	61.292	12.101
Evening	18.993	15.697	0.289	66.776	13.510
Morning	17.130	13.819	0.317	66.111	12.852
Night	22.172	18.498	0.384	68.752	15.555

Metric: Wind Penetration

Summary statistics by Demand Period:

Demand Period	Mean	Median	Min	Max	Std Dev
Evening_Peak	0.053	0.052	0.015	0.122	0.013
Morning_Peak	0.053	0.052	0.016	0.158	0.013
Off_Peak	0.049	0.048	0.012	0.133	0.014
Standard	0.052	0.051	0.013	0.127	0.013

Metric: Mean Abs Deviation

Demand Period	Mean	Median	Min	Max	Std Dev
Evening_Peak	0.156	0.154	0.060	1.213	0.036
Morning_Peak	0.155	0.154	0.061	0.426	0.035
Off_Peak	0.140	0.137	0.037	0.423	0.036
Standard	0.153	0.151	0.013	0.725	0.035

Metric: Max Abs Deviation

Demand Period	Mean	Median	Min	Max	Std Dev
Evening_Peak	0.063	0.000	0.000	5.733	0.309
Morning_Peak	0.062	0.000	0.000	10.683	0.320
Off_Peak	0.043	0.000	0.000	9.683	0.294
Standard	0.056	0.000	0.000	11.717	0.311

Metric: Minutes Outside Normal

Demand Period	Mean	Median	Min	Max	Std Dev
Evening_Peak	16.936	14.040	0.289	60.372	11.765
Morning_Peak	17.606	14.109	0.317	66.111	13.325
Off_Peak	22.123	18.465	0.359	68.752	15.496
Standard	17.633	14.513	0.301	64.626	12.743

Metric: Wind Penetration

Summary statistics by Is Weekend:

Is Weekend	Mean	Median	Min	Max	Std Dev
0	0.052	0.051	0.012	0.158	0.014
1	0.050	0.049	0.013	0.122	0.014

Metric: Mean Abs Deviation

Is Weekend	Mean	Median	Min	Max	Std Dev
0	0.151	0.149	0.037	1.213	0.037
1	0.144	0.143	0.013	0.426	0.035

Metric: Max Abs Deviation

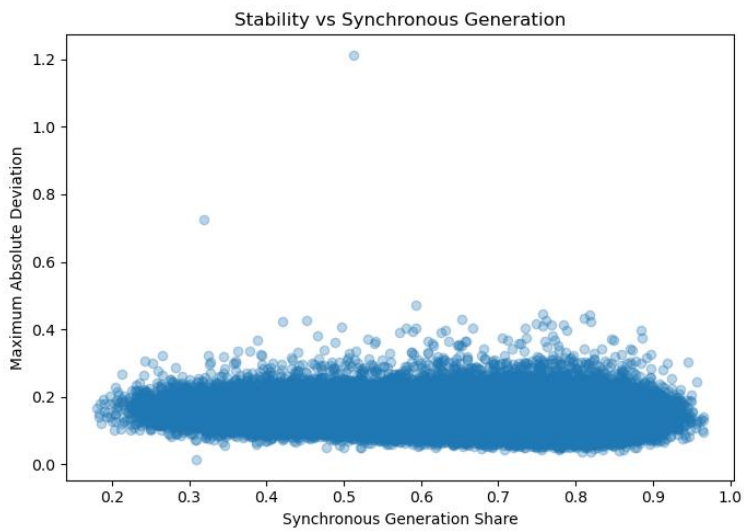
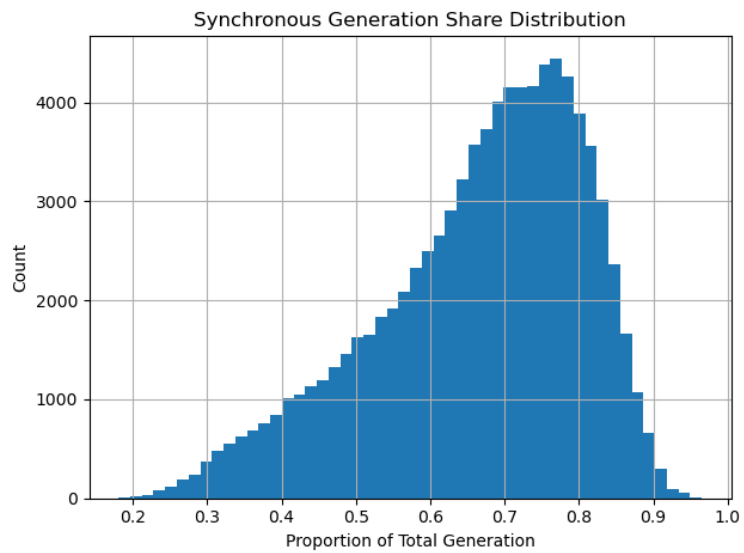
Is Weekend	Mean	Median	Min	Max	Std Dev
0	0.059	0.000	0.000	11.517	0.322
1	0.038	0.000	0.000	11.717	0.262

Metric: Minutes Outside Normal

Is Weekend	Mean	Median	Min	Max	Std Dev
0	18.393	15.134	0.289	68.752	13.420
1	20.548	17.066	0.485	68.622	14.672

Metric: Wind Penetration

1.2.4 Synchronous Generation Analysis



1.2.5 Storm Events Analysis

Total storm events: 1004 hours (1.14% of data)

Metric	Normal Conditions	Storm Conditions	Difference (%)
Mean Abs Deviation	0.051	0.052	0.9%
Max Abs Deviation	0.149	0.150	0.2%
Minutes Outside Normal	0.053	0.073	37.4%

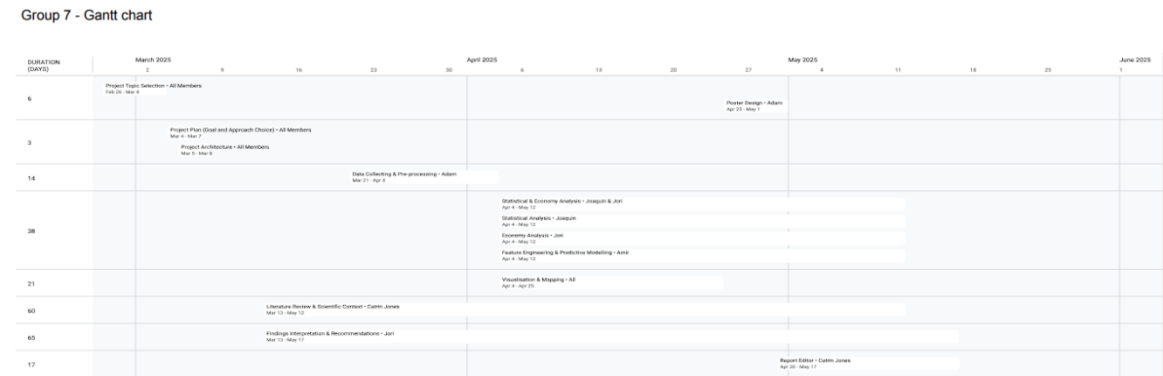
Appendix 2 - Methodology and Planning

2.1 Planning

Gantt Chart

Group 7 - Gantt Chart								
PROJECT TITLE			Assessing the Impact of Wind Renewable Energy on UK Grid Stability					
PROJECT MANAGER			Henok Mekuria					
WBS NUMBER	TASK TITLE	TASK OWNER	SUPPORTER	DEPENDENCIES	START DATE	DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE
1	Project Definition and Planning							
1.1	Project Topic Selection	All Members			26/02/25	04/03/25	6	100%
1.2	Project Plan (Goal and Approach Choice)	All Members		1.1	04/03/25	07/03/25	3	0%
1.3	Project Architecture	All Members		1.2	05/03/25	08/03/25	3	0%
2	Project Execution and Monitoring							
2.1	Data Collecting & Pre-processing	Adam	Henok	1	21/03/25	04/04/25	14	0%
2.2	Statistical & Economy Analysis	Joaquin & Jori		2.1	04/04/25	12/05/25	38	0%
2.2.1	Statistical Analysis	Joaquin		2.1	04/04/25	12/05/25	38	0%
2.2.2	Economy Analysis	Jori		2.2.1	04/04/25	12/05/25	38	0%
2.3	Feature Engineering & Predictive Modelling	Amir	Henok	2.2	04/04/25	12/05/25	38	0%
2.4	Visualisation & Mapping	All		2.2	04/04/25	25/04/25	21	0%
2.5	Literature Review & Scientific Context	Catrin Jones		1.1; 1.2	13/03/25	12/05/25	60	0%
2.6	Findings Interpretation & Recommendations	Jori		2.2; 2.3; 2.5	13/03/25	17/05/25	65	0%
2.7	Poster Design	Adam	Jori	2.4; 2.6	25/04/25	01/05/25	6	0%
2.8	Report Editor	Catrin Jones	Jori	2.1; 2.2; 2.3; 2.4; 2.5; 2.6	30/04/25	17/05/25	17	0%

Timeline

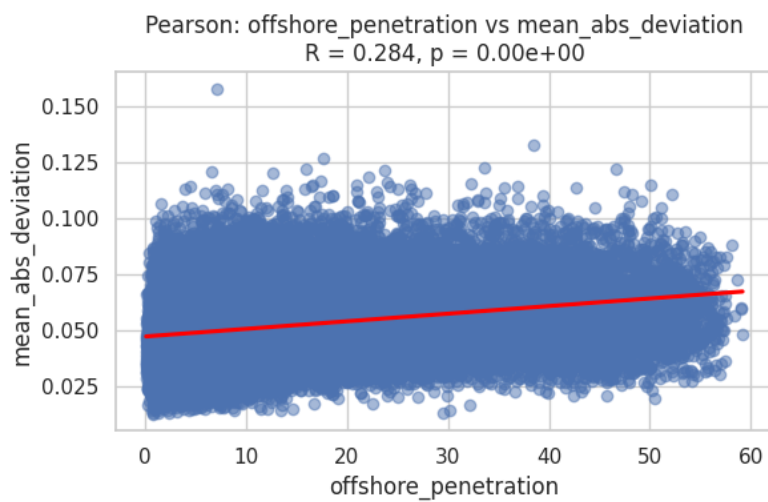
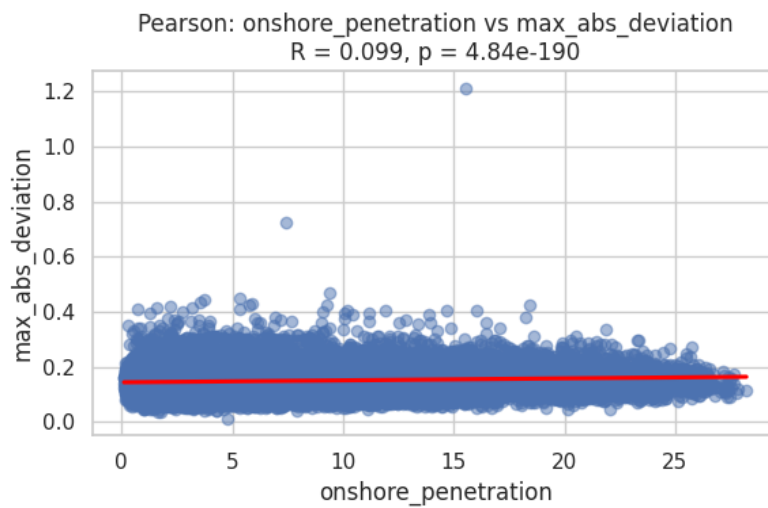
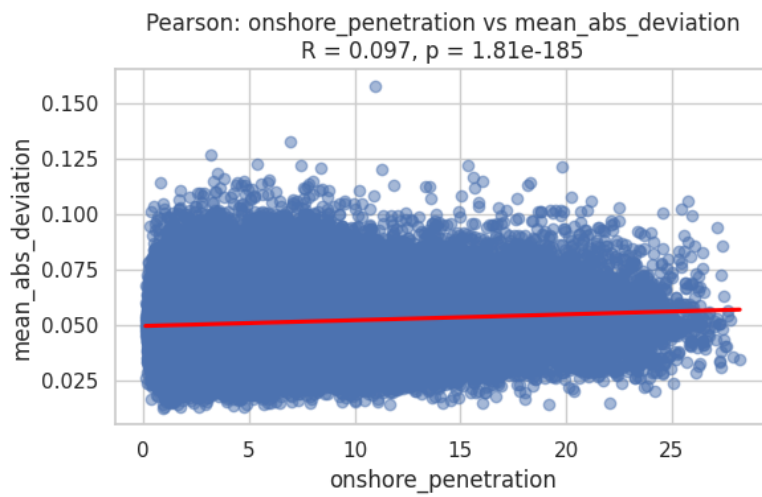


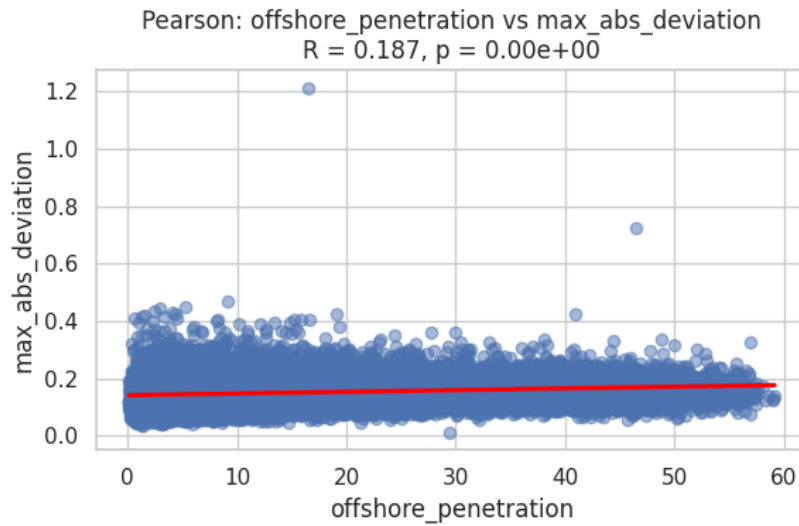
2.2 Research Question 1

2.2.1 Correlation Analysis

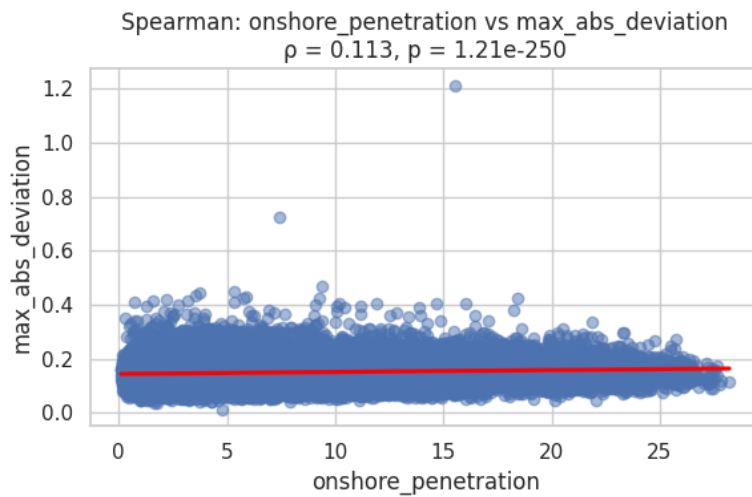
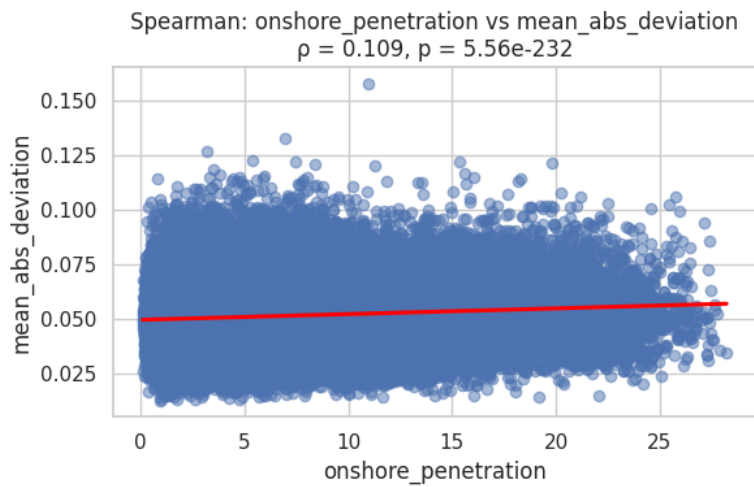
Penetration Type	Metric	Pearson r	Pearson p	Spearman ρ	Spearman p
Onshore	mean_abs_deviation	0.097	1.81e-185	0.109	5.56e-232
Offshore	mean_abs_deviation	0.284	0	0.324	0
Onshore	max_abs_deviation	0.099	4.84e-190	0.113	1.21e-250
Offshore	max_abs_deviation	0.187	0	0.236	0

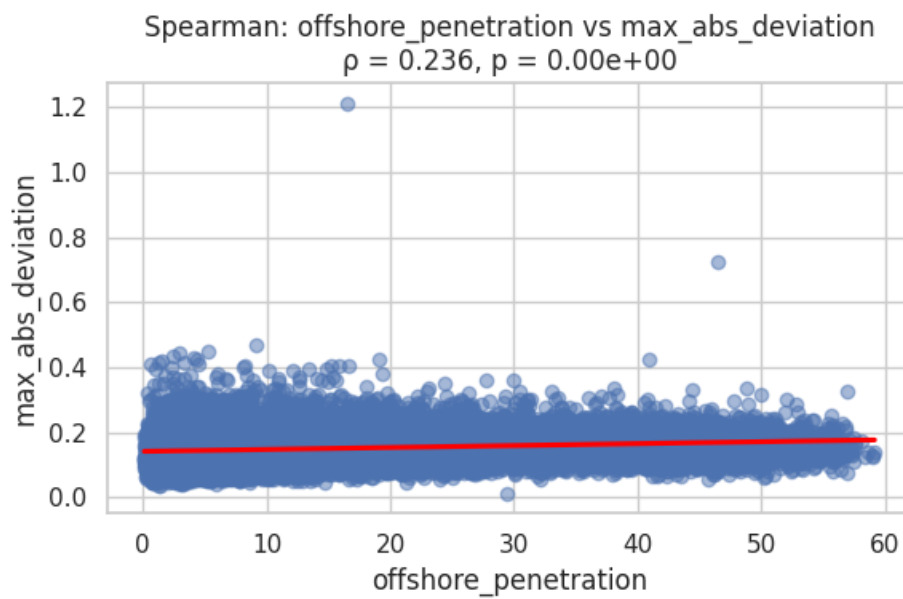
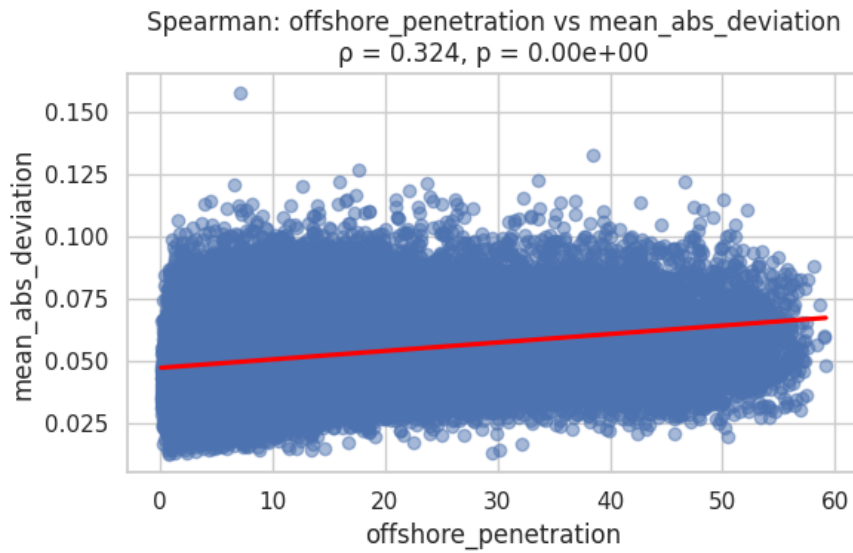
2.2.2 Pearson Correlation Graphs





2.2.3 Spearman Correlation Graphs





2.2.4 Onshore Penetration – Binned Statistics

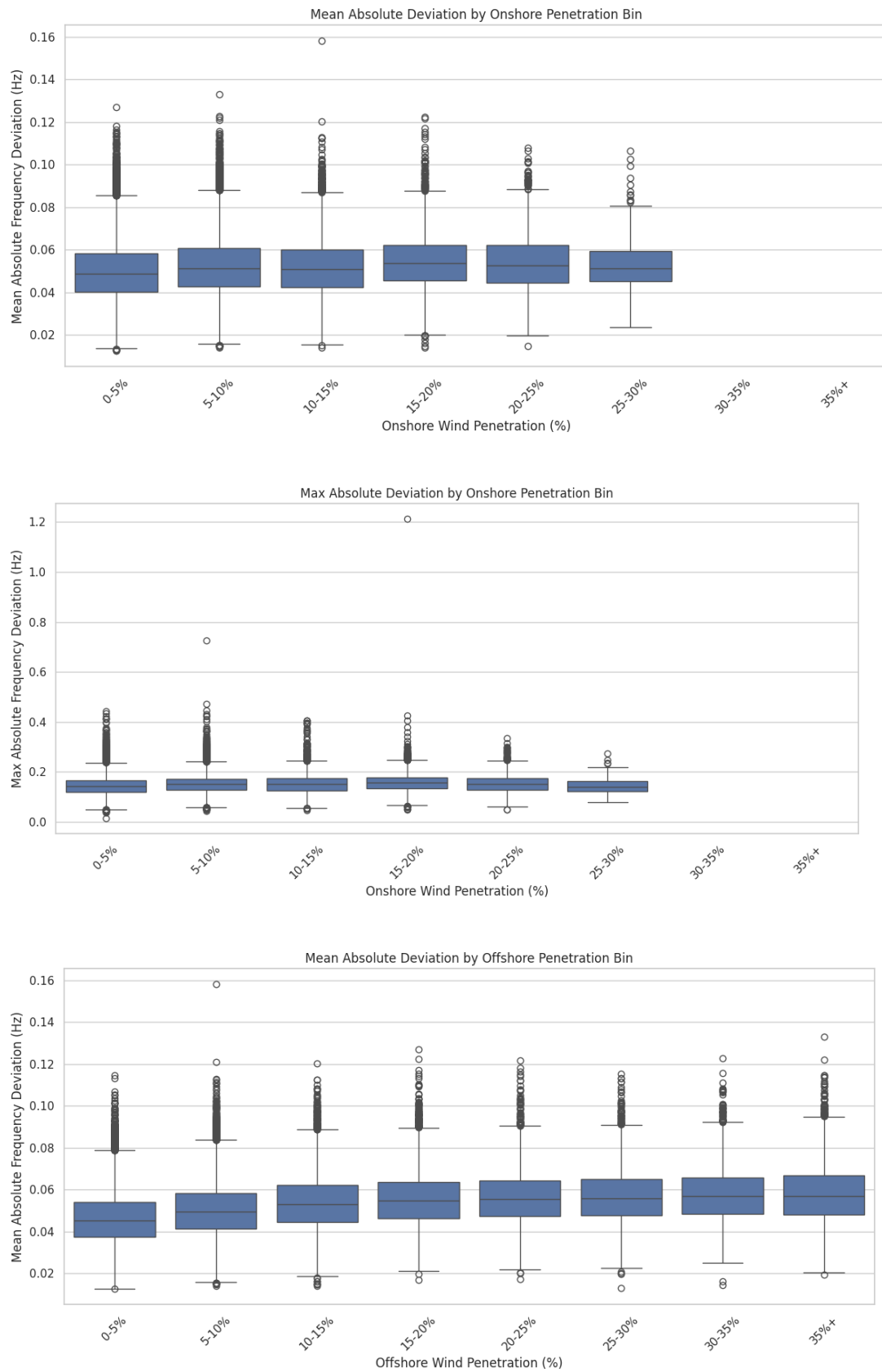
Bin	mean_abs_dev (mean)	median	std	max_abs_dev (mean)	median	std
0–5%	0.0499	0.0489	0.0135	0.1453	0.143	0.0356
5–10%	0.0523	0.0514	0.0137	0.1517	0.150	0.0359
10–15%	0.0517	0.0509	0.0134	0.1516	0.150	0.0370

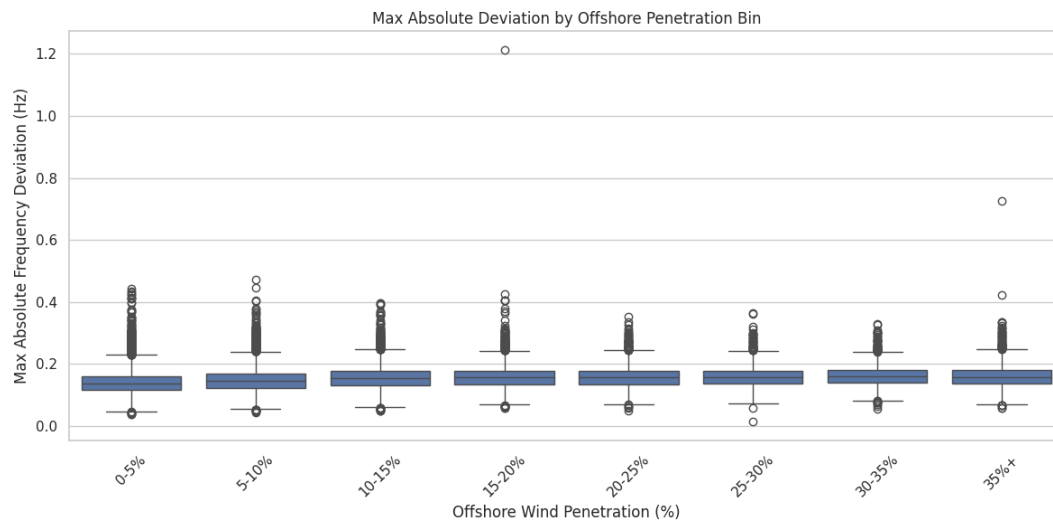
15–20%	0.0545	0.0537	0.0132	0.1576	0.156	0.0380
20–25%	0.0538	0.0528	0.0136	0.1542	0.152	0.0366
25–30%	0.0542	0.0511	0.0149	0.1451	0.141	0.0335
30–35%	NaN	NaN	NaN	NaN	NaN	NaN
35%+	NaN	NaN	NaN	NaN	NaN	NaN

2.2.5 Offshore Penetration – Binned Statistics

Bin	mean_abs_dev (mean)	median	std	max_abs_dev (mean)	median	std
0–5%	0.0462	0.0453	0.0123	0.1394	0.137	0.0358
5–10%	0.0502	0.0494	0.0129	0.1476	0.145	0.0354
10–15%	0.0539	0.0531	0.0132	0.1548	0.153	0.0359
15–20%	0.0557	0.0547	0.0131	0.1579	0.156	0.0363
20–25%	0.0566	0.0556	0.0132	0.1580	0.156	0.0336
25–30%	0.0569	0.0560	0.0137	0.1588	0.157	0.0334
30–35%	0.0578	0.0567	0.0136	0.1611	0.159	0.0331
35%+	0.0580	0.0568	0.0143	0.1599	0.158	0.0347

2.2.6 Binned Means Boxplots





2.2.7 Normality Tests (Shapiro-Wilk)

Tested Variable	W	p-value	Normal?
mean_abs_deviation ~ C(onshore_bin)	0.9873	4.80e-61	No
max_abs_deviation ~ C(onshore_bin)	0.9729	3.75e-77	No
mean_abs_deviation ~ C(offshore_bin)	0.9878	3.47e-60	No
max_abs_deviation ~ C(offshore_bin)	0.9697	1.06e-79	No

2.2.8 Levene's Test for Homogeneity of Variance

Test	W	p-value	Equal Variances?
onshore_bin × mean_abs_deviation	3.6247	0.0028	No
onshore_bin × max_abs_deviation	4.3033	0.0006	No
offshore_bin × mean_abs_deviation	29.7917	1.99e-41	No
offshore_bin × max_abs_deviation	11.0181	5.38e-14	No

2.2.9 OLS Coefficients – Standard Errors

Variable	Coefficient	p-value
Intercept	0.0480	0.000
C(season)[Spring]	0.0008	0.000
C(season)[Summer]	-0.0003	0.019
C(season)[Winter]	0.0006	0.000
C(time_of_day)[Evening]	-0.0011	0.000
C(time_of_day)[Morning]	0.0001	0.336
C(time_of_day)[Night]	-0.0053	0.000
onshore_penetration	0.0007	0.000
offshore_penetration	0.0006	0.000
onshore × offshore	-2.76e-05	0.000
wind_speed_10m	-0.0009	0.000
is_weekend	-0.0019	0.000

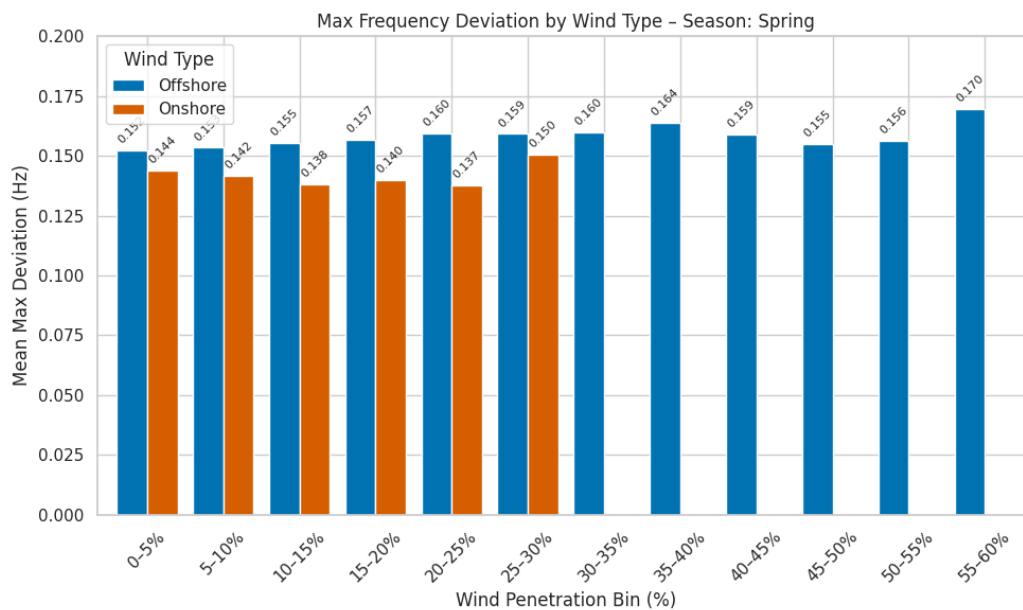
2.2.10 Variance Inflation Factor (VIF)

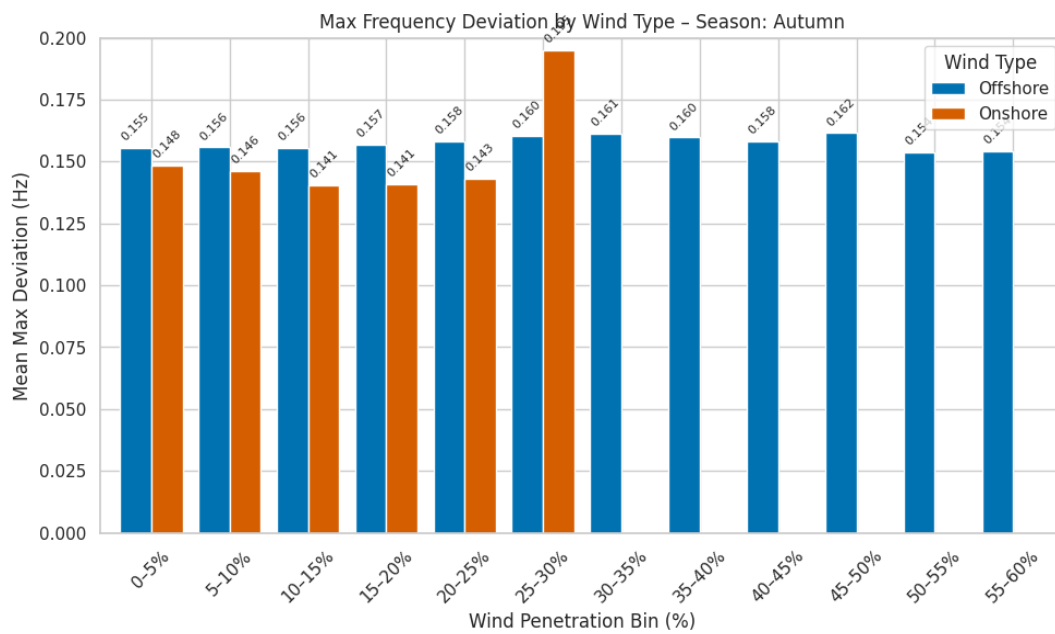
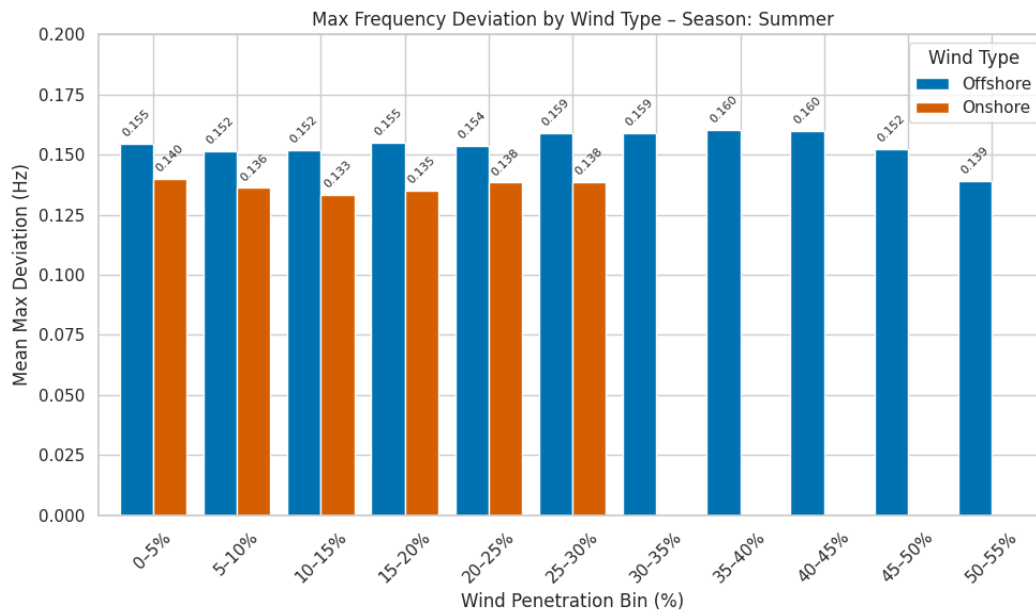
Variable	VIF
onshore_penetration	2.1
offshore_penetration	2.3
onshore × offshore	4.7
wind_speed_10m	1.4
C(season)	1.2
C(time_of_day)	1.8
is_weekend	1.1

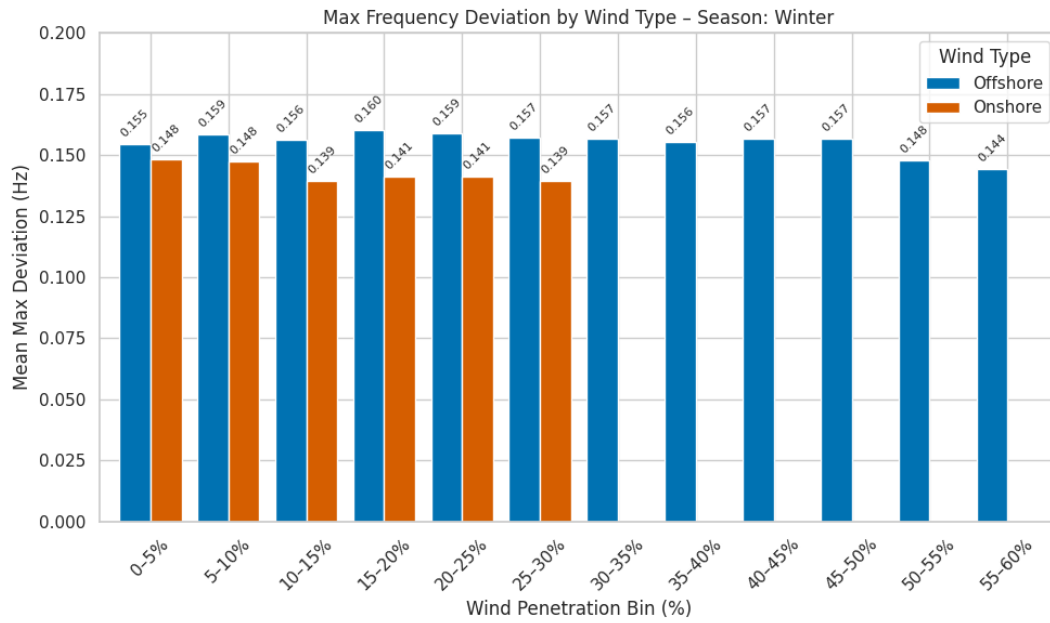
2.2.11 OLS Coefficients – HC3 Robust Errors

Variable	Coefficient	p-value
Intercept	0.0480	0.000
C(season)[Spring]	0.0008	0.000
C(season)[Summer]	-0.0003	0.019
C(season)[Winter]	0.0006	0.000
C(time_of_day)[Evening]	-0.0011	0.000
C(time_of_day)[Morning]	0.0001	0.336
C(time_of_day)[Night]	-0.0053	0.000
onshore_penetration	0.0007	0.000
offshore_penetration	0.0006	0.000
onshore × offshore	-2.76e-05	0.000
wind_speed_10m	-0.0009	0.000
is_weekend	-0.0019	0.000

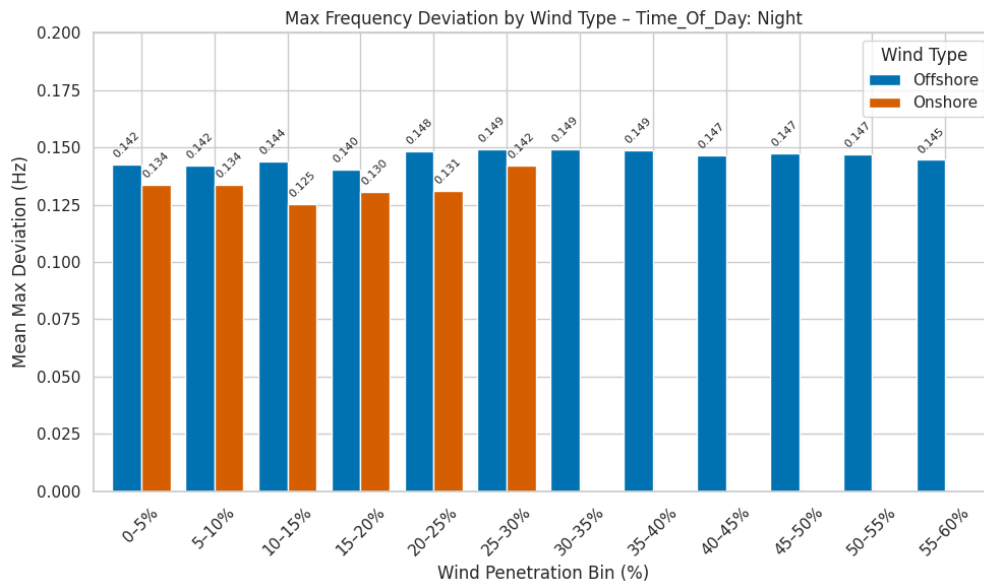
2.2.12 Max Deviation Distribution - Seasonal

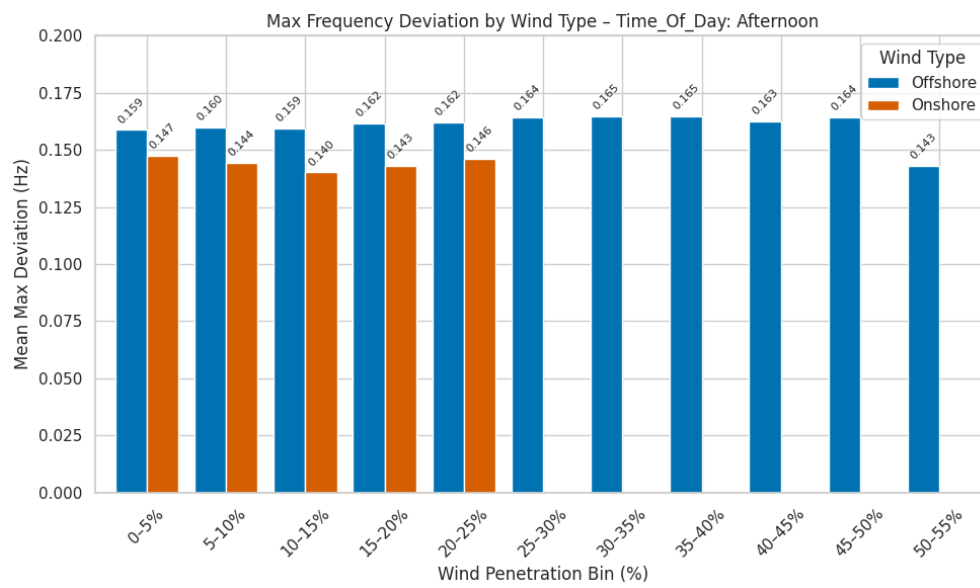
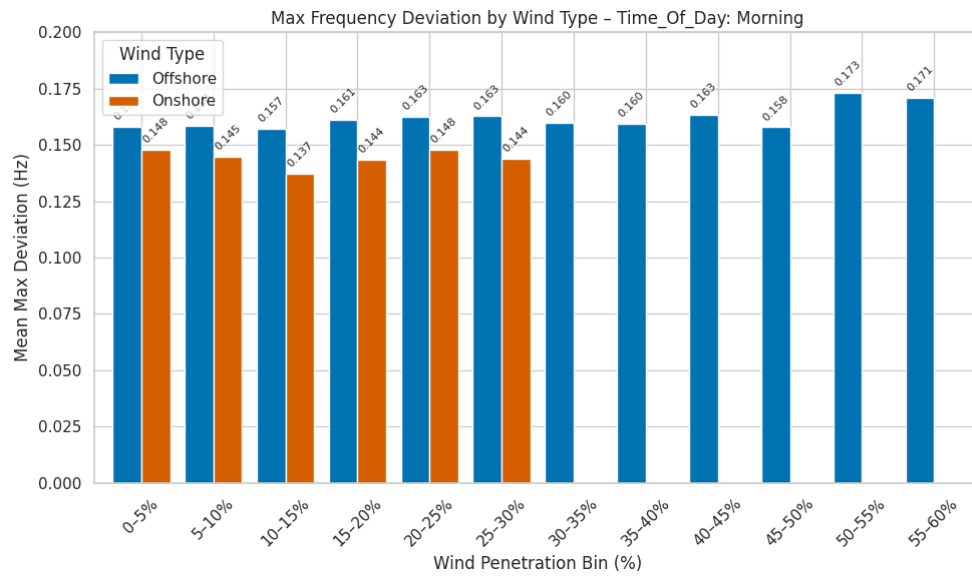


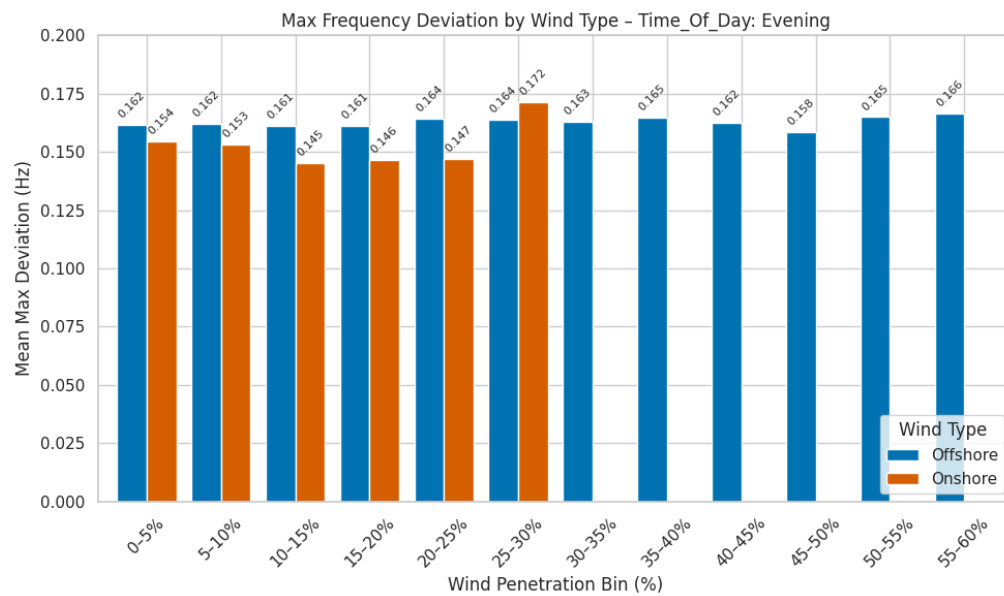




2.2.13 Max Deviation Distribution - Time of Day



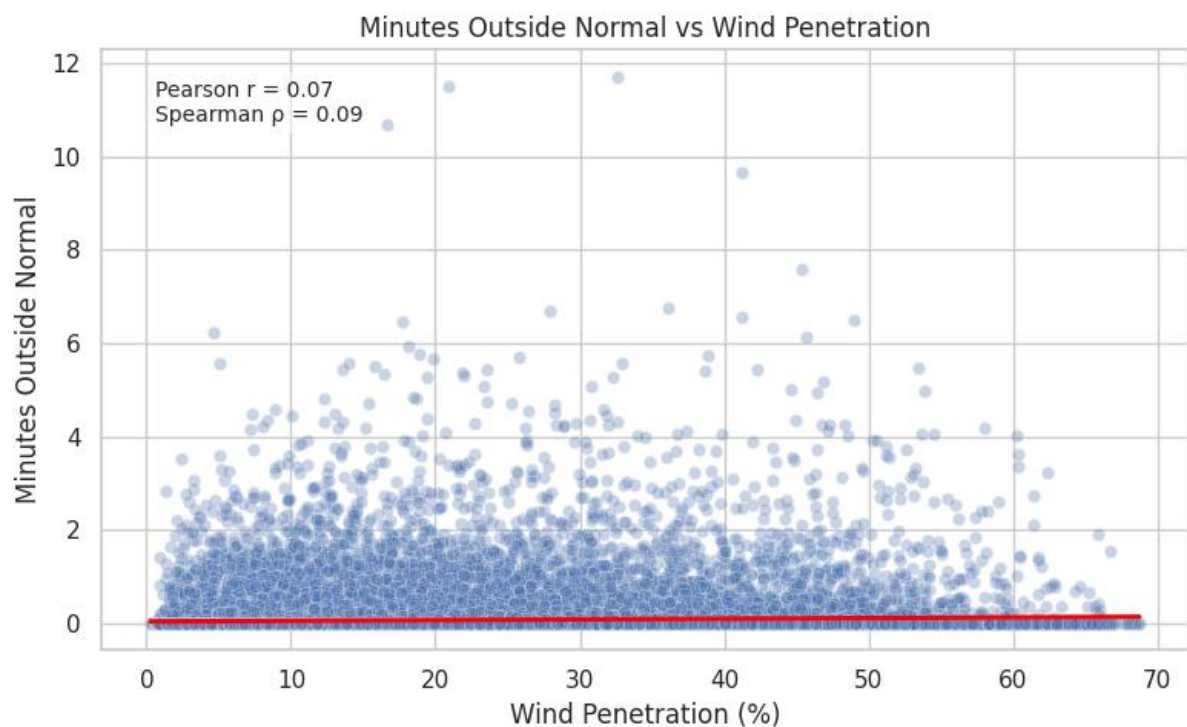




2.3 Research Question 2

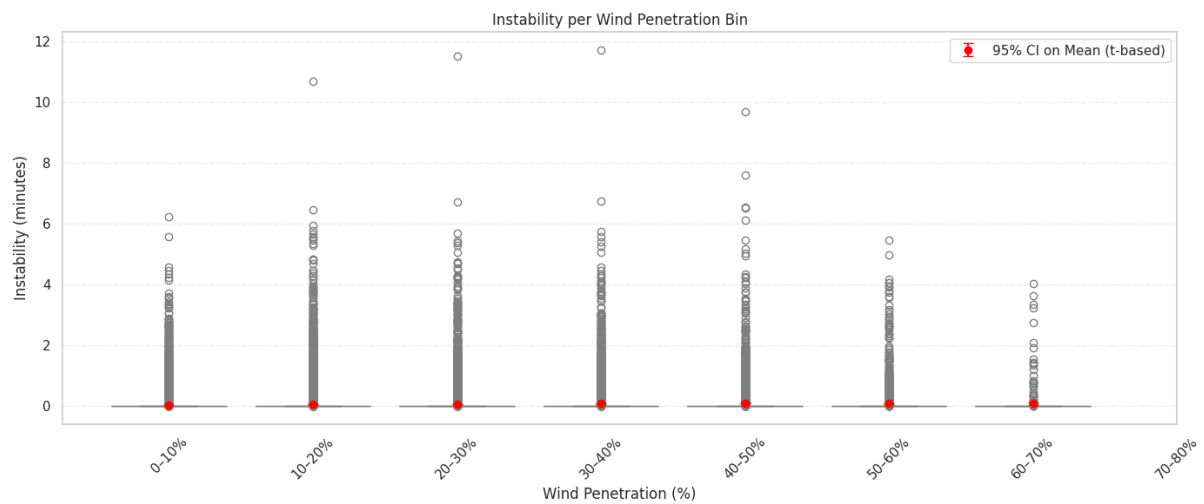
2.3.1 Correlation Analysis

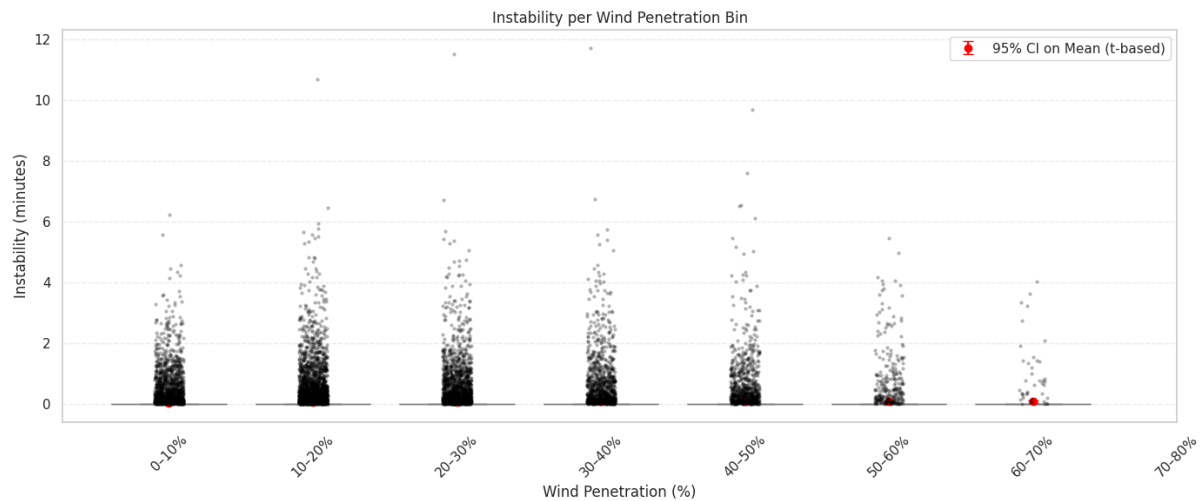
Metric	Pearson r	p-value	Spearman ρ	p-value
Wind Penetration vs. Instability	0.066	7.37e-85	0.090	8.02e-158



2.3.2 Descriptive Stats by Penetration Bin

Penetration Bin	Count	Mean Instability	Std Dev	Max Instability (mins)
0–10%	28,488	0.029	0.202	6.23
10–20%	26,342	0.050	0.296	10.68
20–30%	15,316	0.066	0.339	11.52
30–40%	9,354	0.080	0.393	11.72
40–50%	5,839	0.092	0.429	9.68
50–60%	2,549	0.085	0.412	5.47
60–70%	528	0.076	0.393	4.03



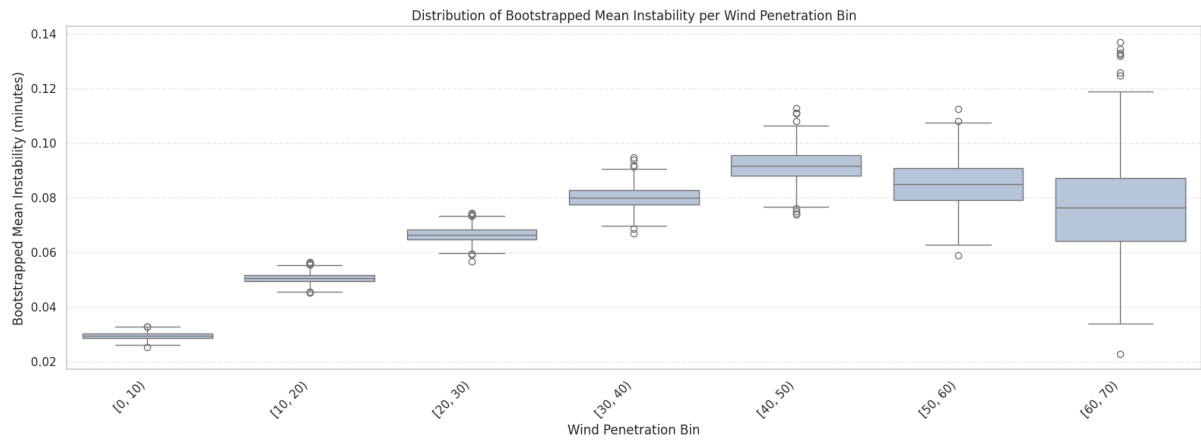


2.3.3 Shapiro Wilk Test for Normality

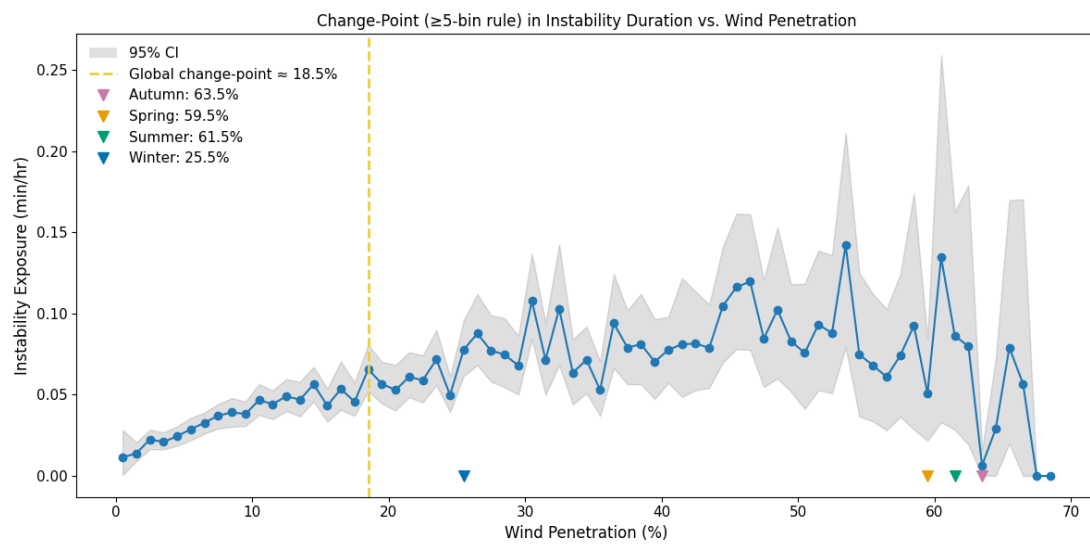
Variable	W Statistic	p-value	Interpretation
minutes_outside_normal	0.1687	1.753×10^{-174}	Strong evidence of non-normality

2.3.4 Bootstrapped Confidence Intervals (95%)

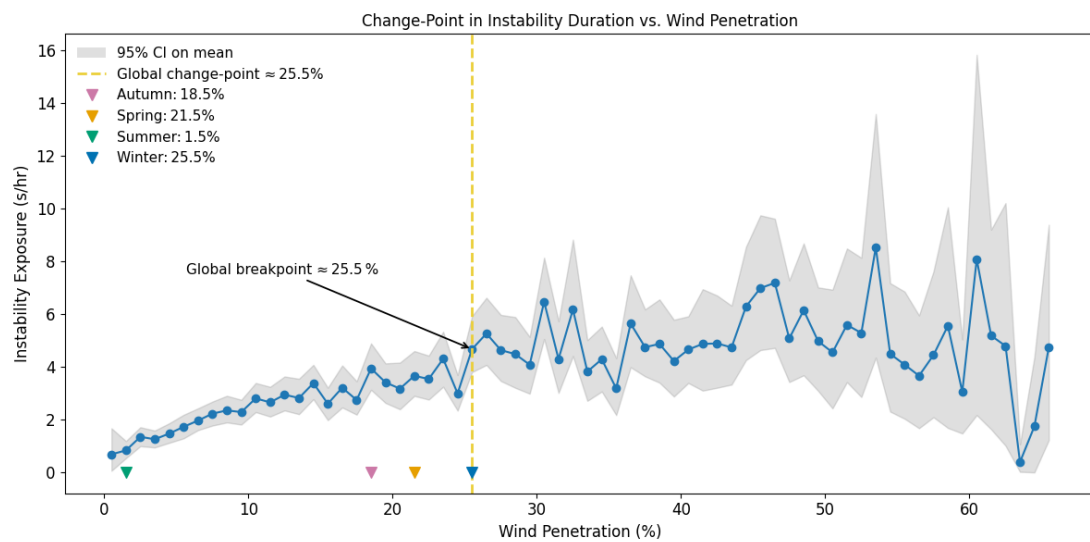
Penetration Bin	Mean	95% CI Lower	95% CI Upper
[0–10%)	0.0293	0.0271	0.0315
[10–20%)	0.0504	0.0470	0.0542
[20–30%)	0.0664	0.0613	0.0720
[30–40%)	0.0797	0.0721	0.0878
[40–50%)	0.0919	0.0806	0.1040
[50–60%)	0.0847	0.0694	0.1005
[60–70%)	0.0762	0.0458	0.1130



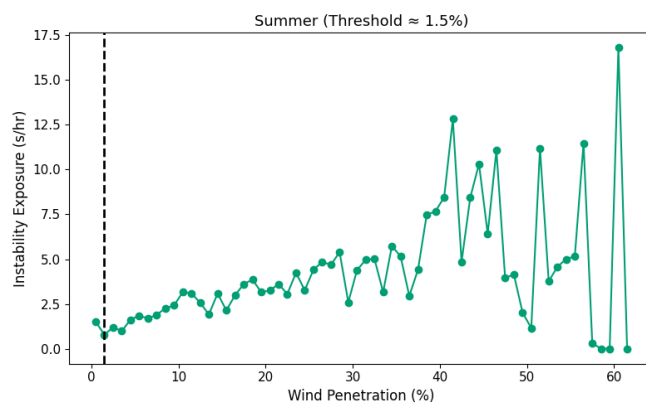
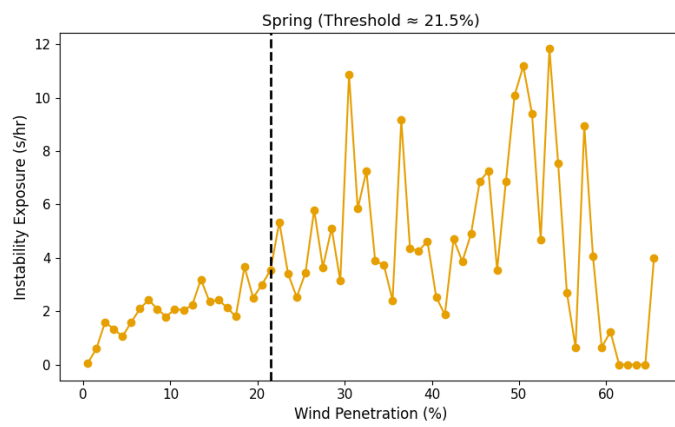
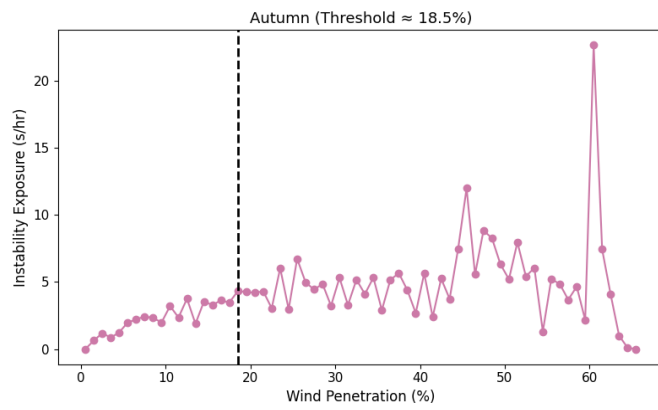
2.3.5 Minimum bins ≥ 5 Constraint

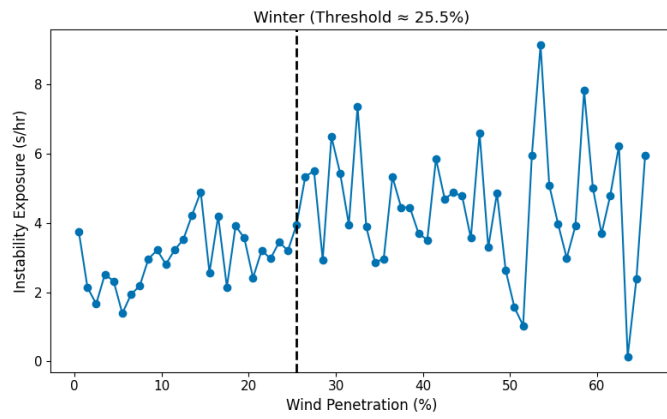


2.3.6 Minimum Observations ≥ 30 Constraint

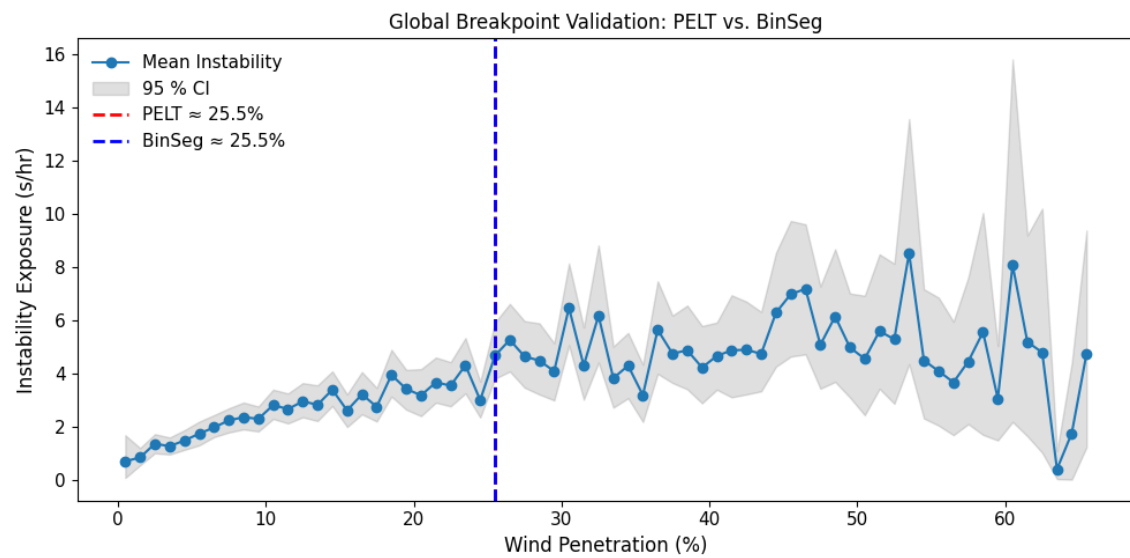


2.3.7 Minimum Observations ≥ 30 Constraint (Seasonal Plots)

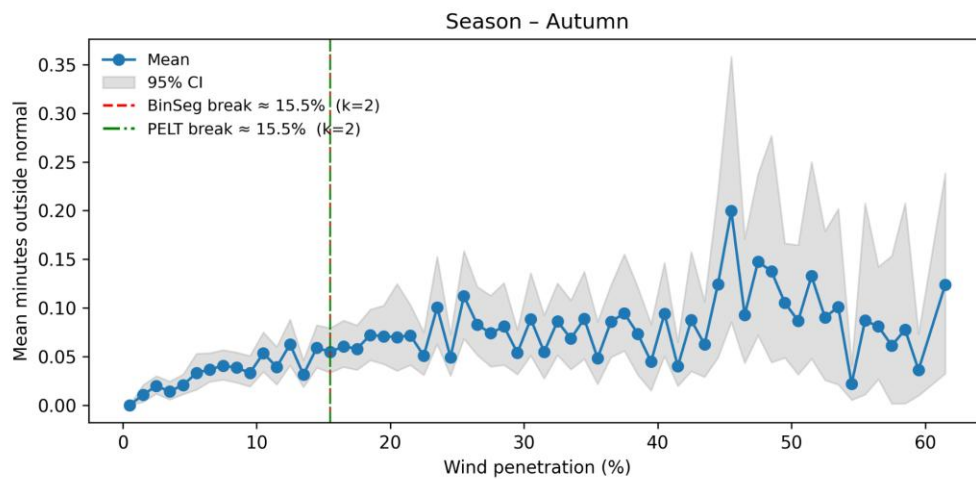


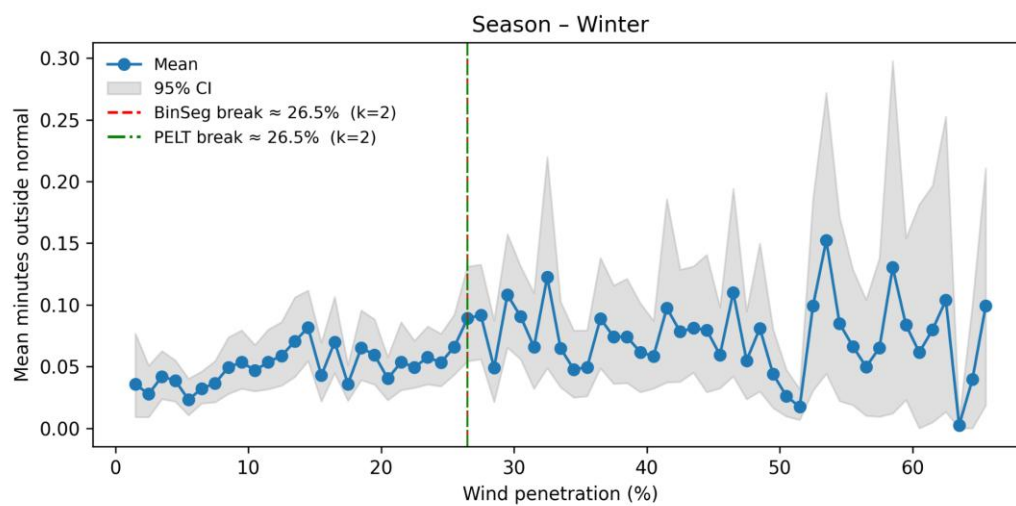
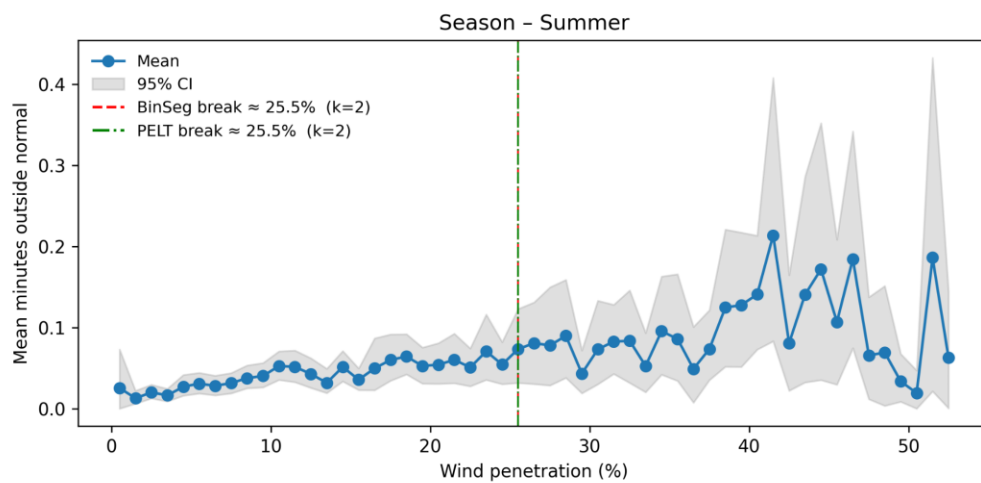
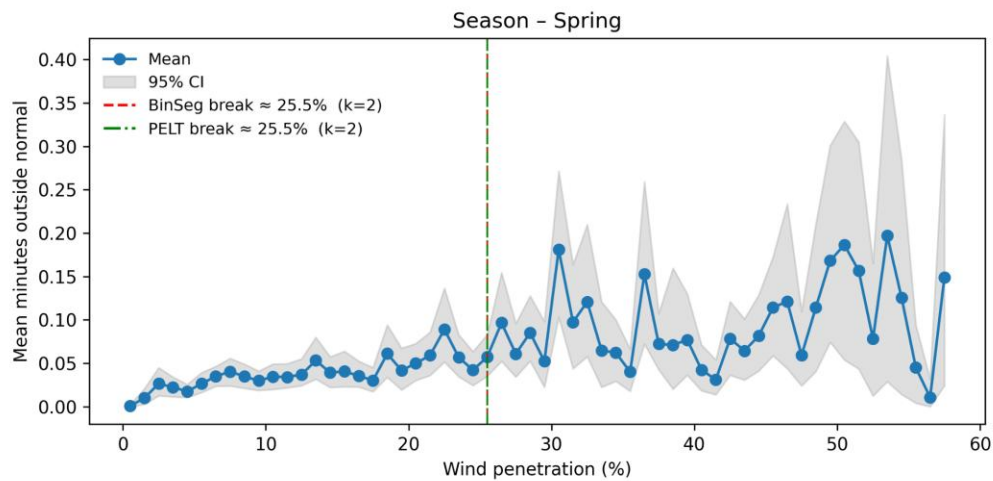


2.3.8 BinSeg vs PELT on $k = 2$ Validation

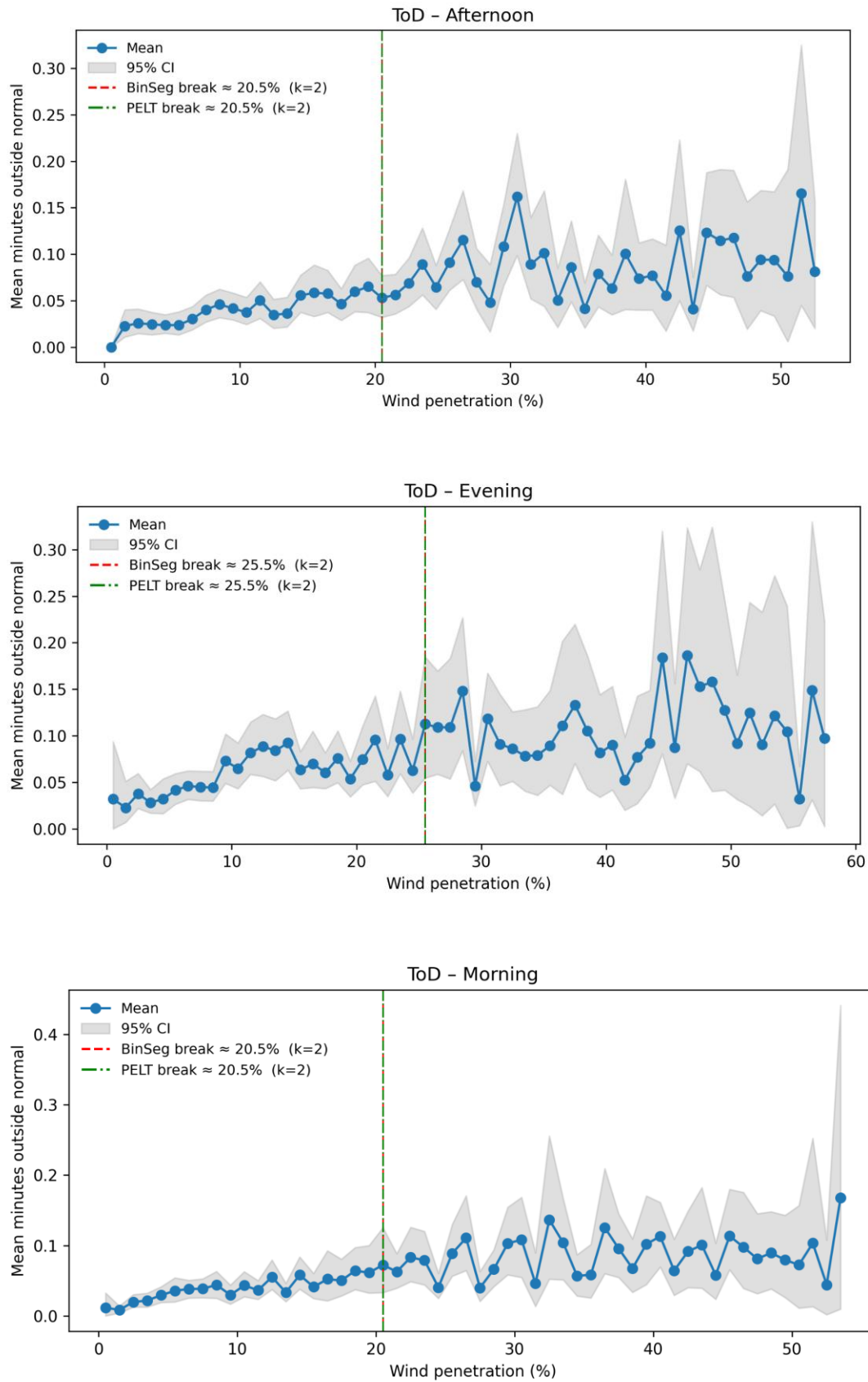


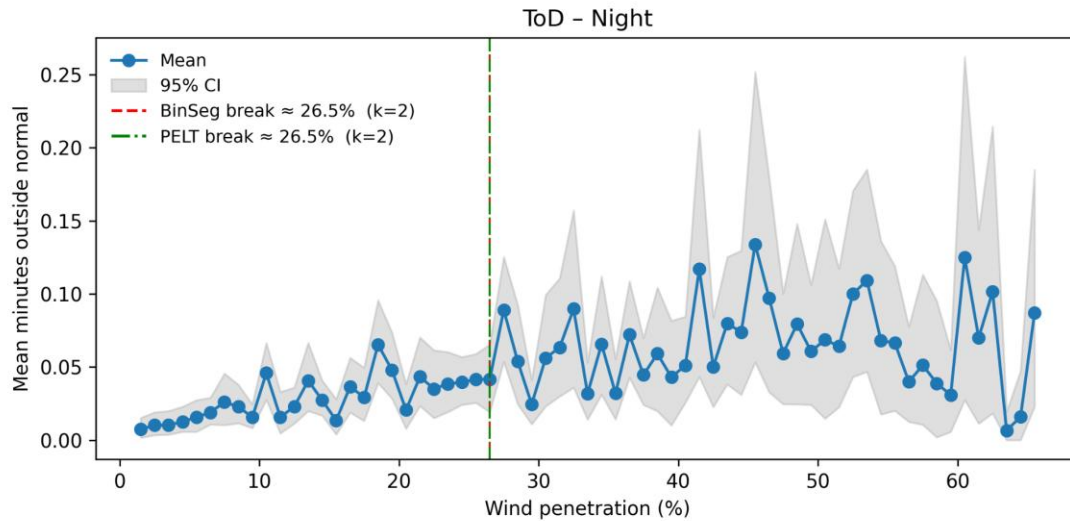
2.3.9 Validated Thresholds - Seasonal Plots





2.3.10 Validated Thresholds - Time of Day Plots





2.4 Research Question 3

2.4.1 Feature Engineering

Feature Group	Features Names	Description
Cyclic time	hour	encode the 24-hour cycle without discontinuities at midnight.
	sin_hr	transform into 24 sin-hour graph
	cos_hr	transform into 24 cos-hour graph
Calendar flags	is_weekend	identifying weekend time
	season	periodic categorical season (Winter/Spring/Summer/Autumn),
	time_of_day	Binning data.
Rolling and trends	wind_roll3	wind_roll3: 3-hour moving average of wind penetration.
	dev_std6	6-hour rolling standard deviation of max deviation. This captures recent volatility and trend.
Delta features	wind_delta	3 hour-to-hour change in wind penetration.
	pressure_trend_3h	Reflect sudden shift in 3-hour pressure difference.
Threshold distances	dist_20	absolute distance of wind penetration from 20 %;
	wind_over_25	capturing excess wind above 25 % (clipped at zero). Models non-linear thresholds.

Polynomial terms	wp,wp2	Second-order terms (wp, wp2) from polynomial expansion of wind_penetration to capture curvature in the wind–instability relationship.
Interaction	wind_x_sync	product of wind_penetration and synch_gen_share (proportion of inertia-providing generation), modelling how wind’s effect depends on available inertia.
Event indicator	highwind_peak	labelling binary flag for wind ≥ 25 % during Evening peak hours (18–24 h), capturing high-stress periods.

2.4.2 Feature selection

1. VIF analysis

Feature name	VIF
wind_roll3	∞
wind_penetration_lag_1	∞
wp	∞
wind_delta	∞
wind_penetration	∞
wind_penetration_lag_2	∞
const (intercept)	652.60
wind_penetration_lag_3	182.92
wp2	171.82
wind_over_25	86.30
wind_x_sync	45.32
synch_gen_share	18.34
dist_20	15.24
time_of_day_Afternoon	8.45
cos_hr	7.34
sin_hr	6.99
time_of_day_Evening	5.52
time_of_day_Morning	5.16
nd	4.61

season_Summer	1.87
season_Winter	1.87
highwind_peak	1.76
season_Spring	1.64
max_abs_deviation_lag_2	1.29
max_abs_deviation_lag_1	1.25
max_abs_deviation_lag_3	1.24
is_weekend	1.20
pressure_trend_3h	1.12
dev_std6	1.09
storm_indicator	1.03

2. Feature selection code snippet

```

selector_proto = SelectFromModel(
    estimator=RandomForestClassifier(
        n_estimators=100,
        class_weight="balanced",
        n_jobs=-1,
        random_state=42
    ),
    threshold="median",           # keep top 50% by importance
    importance_getter="feature_importances_"
)

```

3. Feature selection frequency across 8 cross-validation folds (the bold ones are the selected features):

Feature	Count	Fraction
cos_hr	8	1.000
dev_std6	8	1.000
nd	8	1.000
pressure_trend_3h	8	1.000
max_abs_deviation_lag_3	8	1.000

synch_gen_share	8	1.000
wind_x_sync	7	0.875
wind_penetration_lag_1	7	0.875
wind_penetration_lag_2	7	0.875
wind_penetration_lag_3	7	0.875
max_abs_deviation_lag_1	7	0.875
max_abs_deviation_lag_2	7	0.875
dist_20	7	0.875
wind_delta	7	0.875
wind_roll3	7	0.875
wind_penetration	6	0.750
wp	4	0.500
wp2	3	0.375
is_weekend	2	0.250
sin_hr	1	0.125
time_of_day	1	0.125
wind_over_25	0	0.000
storm_indicator	0	0.000
highwind_peak	0	0.000
season	0	0.000

2.4.3 Model Development Hyperparameters

Model	class_weight / class_weights	max_iter	early_stopping/ early_stopping_rounds	random_state
LogisticRegression	balanced	1000	–	42
RandomForestClassifier	balanced	–	–	42
HistGradientBoostingClassifier	balanced	–	True	42
CatBoostClassifier	class_weights=[1 .0, y_neg/y_pos]	–	early_stopping_rounds =50	42
LightGBMClassifier	balanced	–	–	42

2.4.4 Model Ensembling

1. Histogram-Based Gradient Boosting + LightGBM Ensembling

```
# 1a) Two-model (HistGB+LGBM) weight search
step = 0.05
grid2 = []
for wh in np.arange(0,1+step,step):
    w1 = 1-wh
    ps, ph = [], []
    for fp in fold_preds:
        ens_s = wh*fp["HistGB"] + w1*fp["LGBM"]
        ps.append(average_precision_score(fp["y"], ens_s))
        v_h = (fp["HistGB"]>=0.5).astype(int)
        v_l = (fp["LGBM"]>=0.5).astype(int)
        ens_h = ((wh*v_h + w1*v_l) >= 0.5).astype(int)
        ph.append(average_precision_score(fp["y"], ens_h))
    grid2.append({"w_hist":wh, "w_lgb":w1, "pr_soft":np.mean(ps), "pr_hard":np.mean(ph)})

g2 = pd.DataFrame(grid2)
b2s = g2.loc[g2.pr_soft.idxmax()]
b2h = g2.loc[g2.pr_hard.idxmax()]

print(f"Hist+LGBM soft: PR={b2s.pr_soft:.3f} @ Hist={b2s.w_hist:.2f}, LGBM={b2s.w_lgb:.2f}")
print(f"Hist+LGBM hard: PR={b2h.pr_hard:.3f} @ Hist={b2h.w_hist:.2f}, LGBM={b2h.w_lgb:.2f}")

# 1b) Two-model stacking (HistGB+LGBM)
meta2 = pd.concat([pd.DataFrame(fp) for fp in fold_preds], ignore_index=True)
X2, y2 = meta2[["HistGB", "LGBM"]], meta2["y"]
meta_clf2 = LogisticRegression(class_weight="balanced", random_state=42)
y2_oof = cross_val_predict(meta_clf2, X2, y2, cv=5, method="predict_proba")[:,1]
pr2 = average_precision_score(y2, y2_oof)
meta_clf2.fit(X2, y2)
w2 = meta_clf2.coef_[0] / meta_clf2.coef_[0].sum()
print(f"Hist+LGBM stack: PR={pr2:.3f} @ Hist={w2[0]:.2f}, LGBM={w2[1]:.2f}")
```

2. Logistic Regression + LightGBM Ensembling

```
# 2a) Two-model (LR+LGBM) weight search
grid_lr_lgb = []
for wr in np.arange(0,1+step,step):
    w1 = 1-wr
    ps, ph = [], []
    for fp in fold_preds:
        ens_s = wr*fp["LR"] + w1*fp["LGBM"]
        ps.append(average_precision_score(fp["y"], ens_s))
        v_r = (fp["LR"]>=0.5).astype(int)
        v_l = (fp["LGBM"]>=0.5).astype(int)
        ens_h = ((wr*v_r + w1*v_l) >= 0.5).astype(int)
        ph.append(average_precision_score(fp["y"], ens_h))
    grid_lr_lgb.append({"w_lr":wr, "w_lgb":w1, "pr_soft":np.mean(ps), "pr_hard":np.mean(ph)})

g_lr_lgb = pd.DataFrame(grid_lr_lgb)
b_lr_lgb_s = g_lr_lgb.loc[g_lr_lgb.pr_soft.idxmax()]
b_lr_lgb_h = g_lr_lgb.loc[g_lr_lgb.pr_hard.idxmax()]

print(f"LR+LGBM soft: PR={b_lr_lgb_s.pr_soft:.3f} @ LR={b_lr_lgb_s.w_lr:.2f}, LGBM={b_lr_lgb_s.w_lgb:.2f}")
print(f"LR+LGBM hard: PR={b_lr_lgb_h.pr_hard:.3f} @ LR={b_lr_lgb_h.w_lr:.2f}, LGBM={b_lr_lgb_h.w_lgb:.2f}")

# 2b) Two-model stacking (LR+LGBM)
meta_lr_lgb = pd.concat([pd.DataFrame(fp) for fp in fold_preds], ignore_index=True)
X_lr_lgb, y_lr_lgb = meta_lr_lgb[["LR", "LGBM"]], meta_lr_lgb["y"]
meta_clf_lr_lgb = LogisticRegression(class_weight="balanced", random_state=42)
y_lr_lgb_oof = cross_val_predict(meta_clf_lr_lgb, X_lr_lgb, y_lr_lgb, cv=5, method="predict_proba")[:,1]
pr_lr_lgb = average_precision_score(y_lr_lgb, y_lr_lgb_oof)
meta_clf_lr_lgb.fit(X_lr_lgb, y_lr_lgb)
w_lr_lgb = meta_clf_lr_lgb.coef_[0] / meta_clf_lr_lgb.coef_[0].sum()
print(f"LR+LGBM stack: PR={pr_lr_lgb:.3f} @ LR={w_lr_lgb[0]:.2f}, LGBM={w_lr_lgb[1]:.2f}")
```

3. Histogram-Based Gradient Boosting + Logistic Regression Ensembling

```
# 3a) Two-model (LR+HistGB) weight search
grid_lr_hist = []
for wr in np.arange(0,1+step,step):
    wh = 1-wr
    ps, ph = [], []
    for fp in fold_preds:
        ens_s = wr*fp["LR"] + wh*fp["HistGB"]
        ps.append(average_precision_score(fp["y"], ens_s))
        v_r = (fp["LR"]>=0.5).astype(int)
        v_h = (fp["HistGB"]>=0.5).astype(int)
        ens_h = ((wr*v_r + wh*v_h) >= 0.5).astype(int)
        ph.append(average_precision_score(fp["y"], ens_h))
    grid_lr_hist.append({"w_lr":wr,"w_hist":wh,"pr_soft":np.mean(ps),"pr_hard":np.mean(ph)})

g_lr_hist = pd.DataFrame(grid_lr_hist)
b_lr_hist_s = g_lr_hist.loc[g_lr_hist.pr_soft.idxmax()]
b_lr_hist_h = g_lr_hist.loc[g_lr_hist.pr_hard.idxmax()]

print(f"LR+HistGB soft: PR={b_lr_hist_s.pr_soft:.3f} @ LR={b_lr_hist_s.w_lr:.2f}, HistGB={b_lr_hist_s.w_hist:.2f}")
print(f"LR+HistGB hard: PR={b_lr_hist_h.pr_hard:.3f} @ LR={b_lr_hist_h.w_lr:.2f}, HistGB={b_lr_hist_h.w_hist:.2f}")

# 3b) Two-model stacking (LR+HistGB)
meta_lr_hist = pd.concat([pd.DataFrame(fp) for fp in fold_preds], ignore_index=True)
X_lr_hist, y_lr_hist = meta_lr_hist[["LR","HistGB"]], meta_lr_hist["y"]
meta_clf_lr_hist = LogisticRegression(class_weight="balanced", random_state=42)
y_lr_hist_oof = cross_val_predict(meta_clf_lr_hist, X_lr_hist, y_lr_hist, cv=5, method="predict_proba")[:,1]
pr_lr_hist = average_precision_score(y_lr_hist, y_lr_hist_oof)
meta_clf_lr_hist.fit(X_lr_hist, y_lr_hist)
w_lr_hist = meta_clf_lr_hist.coef_[0] / meta_clf_lr_hist.coef_[0].sum()
print(f"LR+HistGB stack: PR={pr_lr_hist:.3f} @ LR={w_lr_hist[0]:.2f}, HistGB={w_lr_hist[1]:.2f}")
```

4. Logistic Regression + Histogram-Based Gradient Boosting + LightGBM Ensembling

```
# 4a) Three-model (LR+HistGB+LGBM) weight search
grid3 = []
for w1 in np.arange(0,1+step,step):
    for wh in np.arange(0,1-w1+step,step):
        wr = 1-w1-wh
        ps, ph = [], []
        for fp in fold_preds:
            ens_s = wr*fp["LR"] + wh*fp["HistGB"] + w1*fp["LGBM"]
            ps.append(average_precision_score(fp["y"], ens_s))
            v_r = (fp["LR"]>=0.5).astype(int)
            v_h = (fp["HistGB"]>=0.5).astype(int)
            v_l = (fp["LGBM"]>=0.5).astype(int)
            ens_h = ((wr*v_r + wh*v_h + w1*v_l) >= 0.5).astype(int)
            ph.append(average_precision_score(fp["y"], ens_h))
        grid3.append({"w_lr":wr,"w_hist":wh,"w_lgb":w1,"pr_soft":np.mean(ps),"pr_hard":np.mean(ph)})

g3 = pd.DataFrame(grid3)
b3s = g3.loc[g3.pr_soft.idxmax()]
b3h = g3.loc[g3.pr_hard.idxmax()]

print(f"LR+Hist+LGBM soft: PR={b3s.pr_soft:.3f} @ LR={b3s.w_lr:.2f}, Hist={b3s.w_hist:.2f}, LGBM={b3s.w_lgb:.2f}")
print(f"LR+Hist+LGBM hard: PR={b3h.pr_hard:.3f} @ LR={b3h.w_lr:.2f}, Hist={b3h.w_hist:.2f}, LGBM={b3h.w_lgb:.2f}")

# 4b) Three-model stacking
meta3 = pd.concat([pd.DataFrame(fp) for fp in fold_preds], ignore_index=True)
X3, y3 = meta3[["LR","HistGB","LGBM"]], meta3["y"]
meta_clf3 = LogisticRegression(class_weight="balanced", random_state=42)
y3_oof = cross_val_predict(meta_clf3, X3, y3, cv=5, method="predict_proba")[:,1]
pr3 = average_precision_score(y3, y3_oof)
meta_clf3.fit(X3, y3)
w3 = meta_clf3.coef_[0] / meta_clf3.coef_[0].sum()
print(f"LR+Hist+LGBM stack: PR={pr3:.3f} @ LR={w3[0]:.2f}, Hist={w3[1]:.2f}, LGBM={w3[2]:.2f}")
```

5. Logistic Regression + Histogram-Based Gradient Boosting + LightGBM + Random Forest + CatBoost Ensembling

```
# 5) All-five-model ensembles
# 5a) Soft vote (equal weights)
ps5 = []
for fp in fold_preds:
    ens5 = 0.2*fp["LR"] + 0.2*fp["RF"] + 0.2*fp["HistGB"] + 0.2*fp["CB"] + 0.2*fp["LGBM"]
    ps5.append(average_precision_score(fp["y"], ens5))
print(f"All5 soft (equal): PR={np.mean(ps5):.3f}")

# 5b) Hard vote (majority)
ph5 = []
for fp in fold_preds:
    votes = (
        (fp["LR"] >=0.5).astype(int) +
        (fp["RF"] >=0.5).astype(int) +
        (fp["HistGB"]>=0.5).astype(int) +
        (fp["CB"] >=0.5).astype(int) +
        (fp["LGBM"]>=0.5).astype(int)
    )
    pred5 = (votes >= 3).astype(int)
    ph5.append(average_precision_score(fp["y"], pred5))
print(f"All5 hard: PR={np.mean(ph5):.3f}")

# 10c) Stacking
meta5 = pd.concat([pd.DataFrame(fp) for fp in fold_preds], ignore_index=True)
X5, y5 = meta5[["LR", "RF", "HistGB", "CB", "LGBM"]], meta5["y"]
meta_clf5 = LogisticRegression(class_weight="balanced", random_state=42)
y5_oof = cross_val_predict(meta_clf5, X5, y5, cv=5, method="predict_proba")[:,1]
pr5 = average_precision_score(y5, y5_oof)
meta_clf5.fit(X5, y5)
w5 = meta_clf5.coef_[0] / meta_clf5.coef_[0].sum()
print(f"All5 stack: PR={pr5:.3f} @ LR={w5[0]:.2f}, RF={w5[1]:.2f}, HistGB={w5[2]:.2f}, CB={w5[3]:.2f}, LGBM=
```

2.4.5 Feature Importance Analysis

